IPD Reference Notes

Jonathan Ma

June 19, 2026

Contents

Part I

Readings & Literature Review

1. Causally Interpretable Meta-Analysis Combining AD and IPD

Main Objective

The paper addresses a major limitation of modern causally interpretable meta-analysis (CIMA). Existing CIMA methods require IPD from both *every trial* and *target population.*

In practice, many studies only provide aggregate data (AD), such as *means, SDs, sample sizes, published treatment effects.*

The paper proposes a method that allows IPD and AD to be used together within the CIMA framework, resulting in **Aggegrate-Matched Synthetic IPD**:

$$\text{IPD studies} + \text{Aggregate-data studies} = \text{AMSIPD}$$

for studies that only report aggregate summaries.

Motivation

Traditional meta-analysis estimates:

$$\hat{\theta} = \sum_{s=1}^{m} w_s \hat{\theta}_s$$

where weights are usually based on precision (inverse variance) or sample size.

The authors argue that these weights have no *causal interpretation*: A large study receives more weight simply because it is large.

However:

- Larger studies are not necessarily more representative of a target population.
- Trial sample sizes are often driven by funding and logistics.
- Precision-weighted averages may correspond to no meaningful population.

The goal of CIMA is instead to **estimate treatment effects in a specific target population**

Traditional Meta-Analysis vs CIMA

Traditional Meta-Analysis

Target estimand:

$$\sum_{s} w_s \theta_s$$

This is a weighted average of study effects and may not correspond to any actual target population.

CIMA

Target estimand:

$$E[Y^a - Y^{a'} \mid S = 0]$$

where:

- $S = 0$ denotes the target population.
- a and a' denote treatments.

This is the average treatment effect in a specific target population.

The paper repeatedly emphasizes that traditional meta-analysis estimates effects in a potentially unknown population, whereas CIMA estimates effects in a clearly defined target population.

Observed Data Structure

For each individual, O_i takes one of three forms:

- Target Population Sample: $(X_i, S_i = 0)$
- IPD Trial: (X_i, A_i, Y_i, S_i) where:
  - X_i = covariates
  - A_i = treatment
  - Y_i = outcome
  - S_i = study label
- Aggregate Only Trial: $(S_i, \bar{X}_S^A, \bar{Y}_S^A)$ including treatment-arm sample sizes (S), covariate summaries (X), outcome summaries (Y) with no individual-level observations available

CIMA Identification Assumptions

The AMSIPD procedure inherits all assumptions from Dahabreh et al.'s CIMA framework.

1. Consistency

Observed outcome equals potential outcome under received treatment: $Y = Y^A$.

No Hawthorne effects or treatment ambiguities.

2. Conditional Exchangeability

Potential outcomes are independent of treatment assignment given covariates and study.

Because trials are randomized: $Y^a \perp A \mid X, S$

3. Positivity of Treatment Assignment

Every individual could receive either treatment: $P(A = a \mid X, S) > 0$

4. Conditional Exchangeability Between Study and Target Population

A key transportability assumption is that for all studies,

$$E[Y^a - Y^{a'} \mid X, S = 0] = E[Y^a - Y^{a'} \mid X, S = s]$$

After conditioning on measured covariates, treatment effects are the same across studies and the target population.

All treatment-effect heterogeneity is assumed captured by measured covariates.

5. Positivity of Trial Participation

Every covariate pattern in the target population must appear in each study.

$$P(S = s \mid X) > 0$$

Target population must overlap study populations. The authors note that *many practical applications violate this assumption.*

Additional AMSIPD Assumptions

The paper introduces stronger assumptions to reconstruct IPD from aggregate summaries.

Shared Covariance Structure

Covariates may have different means across studies: $E[X_s] \neq E[X_t]$

But covariance matrices are assumed identical $\Sigma_s = \Sigma_t$ for all studies.

Shared Outcome Model

Outcome generation follows $Y = h(A, X) + \epsilon$ with:

- same functional form across studies
- same treatment-covariate relationship

This assumption is crucial because outcome generation for aggregate-only studies relies entirely on relationships learned from IPD studies.

AMSIPD Construction

The procedure consists of two major stages:

Step 1: Generate Aggregate-Matched Covariates

The goal is to reconstruct X_{AMSIPD} for aggregate-only studies.

Homogeneous Version

Assume all IPD studies and the target population share a common joint covariate distribution.

Estimate $f(X)$ using nonparametric density estimation. The paper uses:

- Gaussian kernels for continuous variables
- Discrete kernels for categorical variables

implemented through the `R` package `np`.

Synthetic individuals are generated by:

1. Sampling an observed individual.
2. Drawing a synthetic observation from the estimated kernel density around that individual.

Heterogeneous Version

Instead of pooling all studies:

1. Estimate a separate density for each study.
2. Randomly select a study.
3. Generate a synthetic individual from that study's density.

The purpose is to preserve between-study heterogeneity while relaxing common covariance assumptions.

Matching Aggregate Constraints

1. Synthetic data must reproduce published summaries.
2. Categorical variables are matched exactly using rejection sampling.
3. Continuous variables are recentered and rescaled to match means and SDs.
4. Treatment assignment is then randomized according to original trial allocation.

Step 2: Generate Aggregate-Matched Outcomes

After generating covariates X_{AMSIPD}, the outcomes are generated.

Fit:

$$Y = \beta_0 + \alpha A + \beta X + \gamma AX + \epsilon$$

using studies with IPD. Then predict synthetic outcomes:

$$\hat{Y}_{\text{AMSIPD}} = \hat{\beta}_0 + \hat{\alpha}A + \hat{\beta}X + \hat{\gamma}AX$$

Alternative models may be used, such as nonparametric regression or random forests.

Variance Constraint Problem

Predicted outcomes may become too variable:

$$Var(\hat{Y}_{\text{AMSIPD}}) > Var(Y_{\text{observed}})$$

Simply rescaling outcomes would destroy learned relationships. Instead, the paper proposes **Quadratic Programming** to fit the outcome model under variance constraints.

The goal is to minimize prediction error while ensuring:

$$Var(\hat{Y}_{\text{AMSIPD}}) \leq Var(Y_{\text{observed}})$$

within every study-treatment arm.

This was the preferred implementation in the simulations.

Multiple Imputation Interpretation

AMSIPD contains randomness, therefore:

- Generate multiple AMSIPD datasets.
- Analyze each separately.
- Combine estimates using *Rubin's rules*.

Rubin's Rules

When multiple synthetic or imputed datasets are generated, each dataset is analyzed separately and the resulting estimates are combined using Rubin's rules.

Suppose M datasets are generated. For dataset m, let $\hat{\theta}_m$ denote the estimate of the target parameter, and let U_m denote its estimated variance.

The pooled estimate is

$$\bar{\theta} = \frac{1}{M}\sum_{m=1}^{M}\hat{\theta}_m.$$

The within- and between-imputation variance are

$$\bar{U} = \frac{1}{M}\sum_{m=1}^{M}U_m \qquad B = \frac{1}{M-1}\sum_{m=1}^{M}(\hat{\theta}_m - \bar{\theta})^2.$$

The total variance is

$$T = \bar{U} + \left(1 + \frac{1}{M}\right) B.$$

with resulting standard error

$$SE(\bar{\theta}) = \sqrt{T}.$$

Rubin's rules combine uncertainty within each reconstructed dataset with uncertainty across reconstructed datasets. In the AMSIPD setting, this allows variation across synthetic IPD reconstructions to contribute to the final standard error.

The paper uses $M = 5$ reconstructions.

Simulation Scenarios

Four overlap structures were considered.

- Good Overlap: target population lies inside the union of study populations
- Target population lies between studies, which requries interpolation and violates positivity
- Minimal Overlap: requires extrapolation and expected to perform poorly
- Aggregate only studies overlap strongly with target population, which tests whether AMSIPD can recover information lost by excluding said studies

Main Findings

Case Study

AMSIPD generally outperformed IPD-only analyses and standard weighting approaches.

Outcome-model estimators performed almost as well as full-IPD analyses.

Important Observation

For weighting estimators, AMSIPD often performed better than even the full-IPD version. The authors attribute this to reduction of extreme weights, implicit shrinkage towards aggregate summaries, and improved positivity behavior.

Key Takeaways

1. Traditional meta-analysis lacks a clear target-population interpretation.
2. CIMA provides causally interpretable target-population effects.
3. Existing CIMA requires complete IPD.
4. AMSIPD extends CIMA to mixed IPD + aggregate-data settings.
5. Synthetic IPD are generated by matching aggregate summaries while borrowing structure from IPD studies.

6. Multiple AMSIPD datasets are treated similarly to multiple imputation.

7. Performance is strongest when overlap assumptions are approximately satisfied.

8. Positivity remains the dominant challenge.

2. Comparing Methods for Estimating PSTE in IPD Meta-Analysis

Main Objective

The paper studies how to estimate patient-specific treatment effects (PSTEs) in an IPD meta-analysis when many baseline covariates are available.

The central question is *Can variable selection or shrinkage methods improve estimation of treatment-covariate interactions and patient-specific treatment effects compared to the standard IPD meta-analysis approach?*

The standard approach in many IPD meta-analyses is to include all treatment-covariate interaction terms in the model. The authors investigate whether this leads to overfitting and whether shrinkage methods provide better performance.

Motivation

IPD meta-analysis provides advantages over aggregate-data meta-analysis because:

- It uses patient-level information.
- It has greater power to detect treatment effect heterogeneity.
- It avoids aggregation bias.
- It allows direct estimation of treatment-covariate interactions.

However, modern clinical studies often collect many baseline covariates. Including all treatment-covariate interactions may increase complexity and variance, lead to overfitting, or reduce precision of PTSE estimates.

Conversely, omitting important interactions can miss clinically important effect modifiers. The paper investigates methods that balance these competing concerns.

One-Stage vs Two-Stage IPD Meta-Analysis

The paper briefly reviews the two major frameworks.

Two-Stage Approach

1. Stage 1: Analyze each study separately.

2. Stage 2: Pool study-specific estimates using standard meta-analysis.

This is a familiar framework and has simpler computation, but is less flexible and less efficient.

One-Stage Approach

All patient-level data are modeled simultaneously and patients remain clustered within studies.

This is more flecible and allows easier modeling of treatment-covariate interactions since information borrowing is stronger.

The paper adopts a one-stage framework throughout.

Covariate Types

The authors distinguish three categories of covariates.

Nuisance Covariates

Covariates unrelated to outcome. Examples are variables with no prognostic value or treatment interaction.

Prognostic Factors

Covariates associated with outcome but not treatment effect modification. These affect baseline risk but not treatment benefit.

Effect Modifiers

Covariates that interact with treatment. These are the primary target of the analysis since effect modifiers determine which patients benefit more or less from treatment.

General Statistical Framework

For patient i in study j:

- t_i denotes treatment assignment.
- x_i denotes baseline covariates.
- y_i denotes outcome.

Continuous and binary outcomes are considered. The primary modeling framework is a generalized linear mixed model (GLMM).

For continuous outcomes:

$$y_{ij} \sim N(\mu_{ij}, \sigma^2)$$

with linear predictor

$$\mu_{ij} = a_j + b_j x_i + c_j x_i^{EM} t_i + d_j t_i.$$

Components:

- a_j: study-specific intercept.
- b_j: main covariate effects.

- c_j: treatment-covariate interactions.
- d_j: treatment effect.

Random effects: $d_j \sim N(\delta, \tau^2) \qquad b_{j,m} \sim N(\beta_m, \tau^2_{\beta,m}) \qquad c_{j,m} \sim N(\gamma_m, \tau^2_{\gamma,m})$

For binary outcomes ($y_{ij} \sim Bernoulli(p_{ij})$) with

$$\log\left(\frac{p_{ij}}{1-p_{ij}}\right) = a_j + b_j x_i + c_j x_i^{EM} t_i + d_j t_i.$$

Methods Compared

Seven approaches are compared:

GLMM-Full

The standard approach with all treatment-covariate interactions are included without selection or shrinkage. The model with the most flexibility and least risk of excluding true effect modifiers but downsides of highest overfitting risk and larger variance

GLMM-Oracle

Includes only the true interactions used in data generation. Only used as a theoretical benchmark and not achievable in practice, reprecenting best possible performance

Stepwise Regression

Uses information criteria to add or remove interaction terms. The paper used backwards elimination. This is the most straightforwards and computationally simple, but has poor reproducibility and stability, along with having selection bias.

LASSO

Uses an $\ell_1 : \sum |\beta_j|$ penalty to regularize and performs variable selection automatically.

Ridge Regression

Uses an $\ell_2 : \sum \beta_j^2$ penalty to shrink coefficients while retaining all variables, and doesn't perform variable selection

Adaptive LASSO

Modification of LASSO that applies variable-specific penalties where large coefficients recieve less shrinkage. Designed to improve variable selection consistency/

Bayesian LASSO

Bayesian analogue of LASSO which integrates shrinkage into bayesian inference with Laplace priors on interaction coefficients.

Stochastic Search Variable Selection (SSVS)

Bayesian model averaging approach. Uses indicator variables λ_m to determine whether each interaction enters the model. Rather than selecting a single model, SSVS averages across many models. Outputs posterior inclusion probabilities.

Simulation Design

The authors conduct extensive simulations with both continuous and binary outcomes considered. Data generation varies according to number of studies, patients, effect modifiers, effect heterogeneity, and outcome type.

Scenarioes inclue few true effect modifiers, many nuisance covariates, large and small sample sizes, and different heterogeneity levels.

Performance Metrics

The primary evaluation metric is

$$MSE = E[(\hat{\tau}(PSTE) - \tau(PSTE))^2]$$

Additional metrics include bias, calibration, and accuracy of treatment effect prediction. The emphasis is on individualized treatment estimation rather than merely average treatment effects.

Simulation Results

The most important finding: Shrinkage methods consistently outperform the standard full model and stepwise regression.

Major observations:

1. Shrinkage methods generally have lower MSE.
2. LASSO, adaptive LASSO, ridge, Bayesian LASSO, and SSVS perform similarly well.
3. Stepwise regression performs poorly.
4. Stepwise regression is often worse than including all interactions.
5. Benefits of shrinkage increase when many candidate interactions exist.
6. Benefits increase when true effect modifiers are sparse.

Real Data Applications

Two IPD meta-analysis datasets are analyzed.

Cardiology Dataset

Compared new-generation drug-eluting to bare-metal stents using several baseline characteristics to estimate treatment-covariate interactions

Psychiatry Dataset

Performed meta-analysis of antidepressant treatments and estimated patient-specific treatment benefits using baseline covariates. The real-data examples largely confirm simulation findings that different shrinkage methods produce similar conclusions and stepwise regression remains unstable.

Discussion

The paper argues that patient-specific treatment effect estimation is fundamentally a high-dimensional prediction problem. Treatment-covariate interactions are difficult to estimate and overfitting is a major concern, and importantly shrinkage is generally preferable to hard variable selection.

The authors specifically recommend to *use shrinkage methods when estimating patient-specific treatment effects in IPD meta-analysis* and *avoid step-wise regression.*

The simulations assume covariate information is complete, the model is correctly specified, and there are no missing outcomes, but doesn't address covariate reconstruction, missing covariate data, or aggregate and survival data. The focus is entirely on variable selection and shrinkage once complete IPD are already available.

Main Takeaways

1. IPD meta-analysis is particularly valuable for estimating treatment effect heterogeneity.
2. Treatment-covariate interactions are difficult to estimate reliably.
3. Including all interactions can lead to overfitting.
4. Shrinkage methods substantially improve patient-specific treatment effect estimation.
5. LASSO, adaptive LASSO, ridge, Bayesian LASSO, and SSVS all perform well.
6. Stepwise regression consistently performs poorly.
7. Bayesian and frequentist shrinkage approaches yield broadly similar conclusions.
8. When many candidate effect modifiers are present, shrinkage should be preferred over conventional variable selection.

3. IPD Meta-Analysis vs AD Meta-Analysis

Main Objective

The paper is a methodology review that systematically compares results obtained from:

- Individual Participant Data Meta-Analysis (IPD-MA)
- Aggregate Data Meta-Analysis (AD-MA)

The goal is to determine:

1. How often IPD-MA and AD-MA produce similar conclusions.
2. How often they disagree.
3. Whether disagreements depend on:
    - effect measures used,
    - trials included,
    - participants included,
    - treatment effect modifier analyses versus main treatment effect analyses.

The review seeks empirical evidence rather than theoretical arguments regarding the value of IPD.

Background

Meta-analysis can be conducted using two different levels of information.

Aggregate Data (AD)

Uses published summaries such as means, SDs, hazard ratios, odds ratios, relative risks, and event counts. Researchers do not have access to patient-level observations.

Individual Participant Data (IPD)

Uses the original observations from every participant enrolled in each study.

Advantages often claimed for IPD include:

- Greater statistical power.
- Consistent outcome definitions and inclusion/exclusion criteria.
- Improved subgroup analyses.
- Better handling of missing data and more flexible time-to-event analyses.
- Improved quality control.

However, IPD-MA requires access to original datasets, dataset harmonization, and greater computational effort. The paper investigates whether these advantages translate into materially different conclusions.

Research Questions

The review asks:

1. Do IPD-MA and AD-MA lead to the same statistical conclusions?
2. If they differ, how large are the differences?
3. Are differences more common when studying treatment effect modifiers?
4. Are differences caused by differences in included studies, participants, or analytic methods?

Search Strategy and Inclusion Criteria

Searches were performed in Cochrane library, MEDLINE, and Embase through January 2016. Studie included had to have randomized trial meta-analyses, direct comparison between IPD-MA and AD-MA, and studies were excluded if sufficient numerical information was unavailable.

Study Characteristics

The review identified 39 eligible studies. Across these studies, 190 IPD versus AD comparisons were extracted. Quality assessment classified studies as no important flaws (29 studies, 74%) and possibly important flaws (10 studies, 26%)

For median comparisons, there were roughly 4 comparisons per study, and both IPD-MA and AD-MA had equal participants (1225) while the trials per IPD-MA were 6, and for AD-MA was 7.

Classification of Comparisons

Comparisons were categorized according to:

Type of Analysis

Main treatment effect: 144 comparisons (76%)

Treatment effect modifier analysis: 46 comparisons (24%)

Data Equivalence

The review examined whether identical trials were used, identical participants were used, and identical effect measures were used. This distinction is important because disagreements may arise from available data or methodology.

Standardization of Effect Measures

Different studies reported different effect measures. To enable comparison, estimates were transformed to a common scale using standardized z-scores. The review then compared direction, magnitude, and significance of effects, supplemented with scatterplots and limits-of-agreement analyses

Primary Outcome: Agreement in Statistical Significance

The primary comparison examined statistical significance agreement in both IPD-MA and AD-MA at the two-sided 5% significance level.

Main Findings

Analysis found that four out of five comparisons led to the same statistical conclusion.

Among the 38 disagreements 28 comparisons showed a significant result using IPD-MA that was not detected by AD-MA and 10 comparisons showed a significant result using AD-MA that was not detected by IPD-MA.

Therefore, when disagreement occurred, IPD-MA was more likely to detect a significant effect. This suggests increased power or improved modeling flexibility in IPD analyses.

Direction of Effect

Even when significance agreed, direction sometimes differed. Among the 152 agreements, 23 comparisons showed differing directions of effect. Thus agreement in significance does not necessarily imply agreement in estimated treatment effect.

Main Treatment Effects vs Effect Modifiers

A key result concerns treatment effect modifier analyses. For main treatment effects, IPD-MA more frequently identified significant effects. For effect modifier analyses, no consistent advantage for either method, and results were substantially more variable.

The authors emphasize that treatment effect modification is more difficult to estimate reliably. This is one of the major advantages often cited for IPD.

Identical Data Comparisons

One of the most important findings is that disagreements remained even when same outcomes, participants, or trials were analyzed. Therefore differences between IPD-MA and AD-MA are not explained solely by differences in available data.

Instead, differences can arise because IPD permits different modeling strategies for mdoeling, missing data, and subgroup analyses.

Limits of Agreement Analysis

The review examined agreement in standardized effect estimates, ratio effect estimates, and standard errors.

Average differences were generally small but limits of agreement were wide, so average agreement alone may mask important discrepancies.

Treatment Effect Modifier Analyses

The paper highlights an important distinction: Main treatment effects are often estimable from aggregate data, however treatment effect modifiers require patient-level variation (Age, Sex, Biomarker).

Aggregate summaries frequently lack sufficient information to estimate these relationships. Consequently IPD may offer larger benefits in these settings. The review notes that empirical evidence remains limited.

Authors' Conclusions

In many cases, IPD-MA and AD-MA lead to similar results and conclusions. However, IPD provides opportunities for additional, more detailed, and potentially more appropriate analyses.

The authors recommend:

1. Conduct an AD-MA first.
2. Evaluate whether important limitations remain.
3. Pursue IPD collection only if additional benefits justify the cost.

Key Takeaways

1. IPD-MA and AD-MA agree approximately 80% of the time.
2. Disagreement occurs in roughly 20% of comparisons.
3. When disagreement occurs, IPD-MA is more likely to detect statistically significant effects.
4. Differences persist even when identical trials and participants are analyzed.
5. Average agreement is high, but limits of agreement are wide.
6. Treatment effect modifier analyses remain one of the strongest motivations for IPD.
7. The primary value of IPD is not simply greater power, but the ability to perform richer and more flexible analyses.
8. Researchers should weigh the additional benefits of IPD against its substantial resource requirements.

4. KM-GPT

Main Objective

The paper introduces KM-GPT, a fully automated pipeline for reconstructing individual patient data (IPD) from Kaplan–Meier (KM) survival plots.

The primary goal is to eliminate the manual steps traditionally required for IPD reconstruction, including curve digitization, axis calibration, risk table transcription, and data extraction.

The authors position KM-GPT as the first fully automated end-to-end system for KM-based IPD reconstruction, combining image processing, LLM reasoning, and iterative survival reconstruction algorithms.

Motivation

Time-to-event outcomes are common throughout clinical trials, epidemiology, and public health. Although Kaplan–Meier curves are routinely published, underlying IPD are rarely released because of patient privacy concerns

As a result, researchers often have access only to limited summary statistics, risk tables, and KM plots, which creates barriers for meta analysis or follow-up studies (subgroup, assumption, etc).

Existing Reconstruction Workflow

Historically, reconstruction involves two stages:

Stage 1: Digitization

Convert visual KM curves into numerical coordinates using software like `DigitizeIt, ScanIt, SurvDigitizeR` which require axis calibration, point selection, and other preprocessing to be done manually.

Stage 2: Survival Reconstruction

After digitization, event times are reconstructed, followed by censoring indicators, and generated pseudo-IPD. The dominant approach is the iKM algorithm of Guyot et al. Subsequent methods, including IPDfromKM, build on this framework.

Limitations of Existing Methods

The authors identify several shortcomings, like human error induced by manual intervention, poor reproducibility, and limited support for non-SVG plots.

Core Contribution

KM-GPT performs figure preprocessing, information extraction, and reconstruction without requiring manual intervention. The key innovation is the use of GPT-5 as a reasoning engine that integrates visual and textual information simultaneously.

Overall Architecture

The system consists of three major modules:

1. Image Processing Unit.
2. Multi-Modality Processing Unit (MMPU).
3. IPD Extraction and Reconstruction Unit.

Image Processing Unit

Purpose: prepare KM figures for downstream extraction. Tasks include: noise removal, figure cleaning, resolution enhancement, region identification. The objective is to standardize heterogeneous KM plots before OCR and reasoning. The paper emphasizes that poor image quality is a major source of reconstruction error.

Multi-Modality Processing Unit (MMPU)

The MMPU is presented as the primary methodological innovation. Its purpose is to transform unstructured visual information into structured data.

$$[\text{KM Plot, Risk Table, OCR}] \rightarrow [\text{Axis Config, Meta-data, Number of Curves}]$$

Stage 1: OCR Extraction

The paper uses different image-processing strategies depending on the region.

Risk Tables

Adaptive Gaussian thresholding is used since risk tables usually have uneven contrast, small fonts, and complex backgrounds.

Axis Labels

Global thresholding is used since axis labels are isolated and high contrast, making them easy to segment.

Stage 2: GPT-5 Multi-Modal Reasoning

The image is encoded and passed to GPT-5 together with OCR-derived text.

GPT-5 performs visual and contextual reasoning, along with cross-model validation. It is tasked with identifying group labels, treatment arms, matching risk tables to curves, resolving overlapping graphical elements, and extracting confidence intervals.

The paper argues that these tasks are difficult using OCR or computer vision alone. GPT-5 provides semantic understanding that traditional approaches lack.

Output of MMPU

The final output is a structured JSON object containing all plot data, and this representation is passed to the reconstruction stage.

IPD Extraction and Reconstruction

After parameter extraction, KM-GPT reconstructs IPD. The workflow contains several stages:

Axis Calibration

Pixel coordinates are transformed into time values and survival probabilities using extracted axis boundaries, which creates a mapping from image space to survival-analysis space.

Curve Isolation

The system identifies pixels belonging to each survival curve through greedy path tracing and K-mediods clustering to recover ordered trajectories

Handling Overlapping Curves

For overlapping curves, kNN classification is applied and confidence scores are assigned, which alleviates ambiguous curve assignments

Survival Reconstruction

Once digitized coordinates are obtained, the Guyot iKM algorithm is applied and risk tables are inforporated, and other survival information is handled accordingly.

The algorithm attempts to ensure risk-set consistency and agreement with published curves and risk tables.

Final Reconstructed Dataset

The reconstructed IPD contains survival time, event indicator, and treatment label, stored in CSV format.

Validation Procedure

To verify reconstruction quality:

1. Reconstructed IPD are generated.
2. KM curves are re-estimated from the reconstructed IPD.
3. Reconstructed KM curves are compared to original KM curves.

This serves as an internal validation step.

Web Interface

The paper describes a public web application which involves a RAG/LMAR agent whose role is restricted to model-specific troubleshooting and support.

Synthetic Data Evaluation

The authors first evaluate KM-GPT using simulated survival datasets.

Three parameters (sample size, median survival time, censoring rate) were varied into three levels, producing 27 parameter combinations. 20 replicated were generated, resulting in 540 total curves to work with.

Synthetic Survival Generation

Survival times were generated from exponential distributions. Right censoring was introduced through randomly assigned censoring times. Observed times were defined by the minimum of event time and censoring times.

Parameter Extraction Results

Out of 540 KM plots, 538 were successfully processed, with the two failures noted as digitization errors. Axis, tick, and risk table were fully accurate.

Comparison with SurvdigitizeR

The authors compare KM-GPT with `SurvdigitizeR`, under the assumption that risk tables are manually removed (limitation of the package). They found that KM-GPT reduced failure rate by 0.29, attreibuted to digitization error. The authors attribute KM-GPT's advantage to its automated preprocessing and multi-modal reasoning.

Reconstruction Accuracy

Reconstructed IPD were compared with ground-truth survival data. The primary metric was absolute error (AE=0.005) in survival probabilities. Note that error variance increased in late follow-up and there was adverse tail behavior, attributed to stochastic reconstruction error.

Integrated Absolute Error

Overall curve fidelity was evaluated using Integrated Absolute Error (IAE=0.018):

$$IAE = \int |S_{\text{truth}}(t) - S_{\text{reconstructed}}(t)|\, dt$$

where lower values are closer agreement.

Key Conclusions

1. KM-GPT is the first fully automated KM-to-IPD reconstruction pipeline.

2. The combination of OCR and GPT-5 reasoning enables extraction from heterogeneous KM plots.
3. Automated risk-table extraction is a major advantage over existing approaches.
4. Reconstruction accuracy remains high across diverse survival settings.
5. Performance substantially exceeds traditional semi-manual digitization tools.
6. Reconstruction errors are concentrated in extreme tail regions where visual information is weakest.
7. The resulting IPD are suitable for downstream survival analysis, subgroup analysis, and evidence synthesis.

5. Meta-Analysis of Reconstructed Survival Data (MARS)

Main Objective

The paper proposes a new aggregate-data meta-analysis framework for time-to-event outcomes called **Meta-Analysis of Reconstructed Survival Data (MARS)**

The primary goal is to perform survival meta-analysis using richer survival information than conventional hazard-ratio-based approaches. Rather than relying exclusively on reported hazard ratios, MARS incorporates:

1. Reconstructed survival data from published Kaplan–Meier curves.
2. Reported hazard ratios.
3. Published survival probabilities at specific time points.

The authors argue that reconstructed survival data provide substantially more information than hazard ratios alone and allow meta-analysis without requiring the proportional hazards assumption.

Motivation

Individual participant data (IPD) meta-analysis is widely regarded as the gold standard for survival meta-analysis.

However, IPD are rarely available, most published studies report only aggregate data, and less than 1% of published meta-analyses include IPD.

Consequently, most evidence synthesis relies on aggregate-data methods.The standard aggregate-data approach uses study-level hazard ratio estimates. This introduces several limitations.

Dependence on the Proportional Hazards Assumption

Conventional meta-analysis assumes: $HR(t) = HR$ for all time.

However, survival curves frequently cross, treatment effects may vary over time, and immunotherapy studies often violate proportional hazards. The authors cite evidence suggesting proportional hazards violations occur in approximately one-fifth of treatment comparisons.

Information Loss

Many studies report Kaplan–Meier curves, survival probabilities, and log-rank statistics, but not hazard ratios.

Traditional approaches either exclude these studies or convert information into hazard ratios using additional assumptions.

Limited Reporting

Hazard ratios alone do not directly describe survival outcomes. The authors argue that clinicians often care more about survival curves than hazard ratios.

Central Idea

The key innovation is to treat reconstructed survival data as an intermediate between AD and IPD. The authors view reconstructed survival data as a rich source of information that can be incorporated directly into a Bayesian hierarchical model. Rather than reducing studies to a single hazard ratio estimate, MARS models reconstructed survival trajectories.

Three Types of Data

MARS combines three different survival-data sources:

- Type 1: Reconstructed (pseudo) survival data obtained from Kaplan–Meier curves
- Type 2: Published hazard ratios after log-transformation along with standard errors
- Type 3: Published survival probabilities at particular time points.

Modeling Framework

MARS uses a Bayesian hierarchical model. The hazard function is modeled using a piecewise exponential structure. Time is partitioned into intervals $I_1, \dots, I_J$ within each interval $h(t) = \lambda_j$ is constant which yields a flexible approximation to arbitrary hazard functions.

Piecewise Hazard Model

For interval $I_j = (t_{j-1}, t_j]$ the hazard is $h(t) = \lambda_j$. The cumulative hazard becomes:

$$H(t) = \sum_{l=1}^{j-1} \lambda_l |I_l| + \lambda_j (t - t_{j-1}).$$

The survival function is:

$$S(t) = \exp[-H(t)].$$

This allows for non-proportional and time variation.

Modeling Treatment Effects

Treatment effects are allowed to vary over time. Instead of $HR(t) = HR$ the model allows:

$$HR(t) = \exp(\beta_j)$$

within each interval. Thus: $\beta_1, \beta_2, \ldots, \beta_J$ represent interval-specific log hazard ratios, which directly relaxes the proportional hazards assumption.

Temporal Smoothing

The interval-specific treatment effects are not estimated independently.

Instead, an AR(1) prior is imposed, ie $Corr(\beta_j, \beta_{j-1}) \neq 0$ which prevents highly unstable interval-specific estimates. The resulting hazard-ratio function changes smoothly over time.

Hierarchical Structure

Study-specific effects are treated hierarchically, so the model accounts for within- and between-differences, creating a random-effects meta-analysis framework. Studies with longer follow-up contribute more information naturally through the likelihood. The authors emphasize that study weighting emerges automatically through the hierarchical model rather than being manually imposed.

Likelihood Contributions

Each data type contributes differently:

- Type 1: Reconstructed survival times contribute through the survival likelihood, having the most information
- Type 2: Reported ratios and variance estimates are incorporated directly into the likelihood
- Type 3: Published survival probabilities contribute through the estimated survival function.

Bayesian Inference

Inference is performed using MCMC. Posterior samples provide HRs, survival probabilities and times, and RMST. Credible intervals are also made available.

Summary Measures

One major advantage of MARS is the ability to estimate clinically interpretable quantities.

The survival function $S(t)$ can be estimated at any time point.

Median survival is defined through $S(t_{MS}) = 0.5$, equivalently $H(t_{MS}) = \log(2)$. Because MARS estimates the cumulative hazard function, median survival is obtained directly.

Restricted Mean Survival Time (RMST)

The paper strongly emphasizes RMST. RMST up to time horizon τ is defined as $RMST(\tau) = E[\min(T, \tau)]$, equivalently:

$$RMST(\tau) = \int_0^\tau S(u)\, du.$$

RMST represents expected survival time accumulated before time horizon τ. This is good since it doesn't need proportional PH to hold and is more interpretable than HRs.

Theoretical Results

The paper establishes asymptotic properties.

Consistency

Under regularity conditions (sample sizes and number of studies increases), the estimators converge to the true parameter values, meaning HR and other survival estimates are consistent.

Relative Efficiency

The paper compares MARS with IPD meta-analysis. MARS is never more efficient than full IPD but approaches IPD asymptotically. As both sample sizes and number of studies grows, the variance of MARS approaches the variance of IPD meta-analysis.

Simulation Study 1: Proportional Hazards

The first simulation assumes proportional hazards. 13 studies with weibull baseline hazard, constant treatment effect, and random censoring are characterized. For data availability, assume some studies have reconstructed data, some have KM curves, and some have survival probabilities. The authors compare MARS, IPD, and AD.

Results

MARS correctly recovered a nearly constant hazard ratio function. Relative efficiency compared to IPD is 1.22 indicating modest efficiency loss.

Simulation Study 2: Non-Proportional Hazards

The second simulation introduces time-varying treatment effects, where the HR changes over time and survival curves eventually cross (intentionally violating PH).

Results

Conventional hazard-ratio meta-analysis performs poorly, while MARS remains accurate.

Acute Myeloid Leukemia Application

The motivating application examines Minimal Residual Disease (MRD) in acute myeloid leukemia. The data consists of 64 studies with 8033 patients, and available information includes reconstructued KM data and published survival curves/hazard ratios.

Main Findings

MRD-negative patients consistently showed better disease-free survival. The estimated 5-year survival for negative and positive was 0.65 and 0.25 respectively, and the RMST was 3.73 and 2.06 years respectively. The application demonstrates how MARS can generate clinically interpretable survival summaries beyond hazard ratios.

Advantages of MARS

The authors highlight several advantages.

1. Uses reconstructed survival data directly.
2. Incorporates multiple forms of survival evidence simultaneously.
3. Relaxes the proportional hazards assumption and produces time-varying hazard-ratio estimates.
4. Estimates survival probabilities, median survival, RMST.
5. Reduces selection bias by allowing inclusion of studies lacking hazard ratios.
6. Approaches IPD efficiency asymptotically.

Limitations

Several limitations are acknowledged.

- Reconstruction quality depends on KM reconstruction accuracy.
- Efficiency remains lower than full IPD.
- Piecewise modeling requires interval specification.
- Time-varying hazard estimation may be sensitive in sparse regions.
- MCMC introduces additional computational burden.

Key Takeaways

1. Reconstructed survival data can serve as an intermediate information source between aggregate data and IPD.
2. MARS integrates reconstructed survival data, hazard ratios, and survival probabilities into a unified Bayesian framework.

3. The method avoids reliance on proportional hazards.

4. Time-varying treatment effects can be estimated directly.

5. RMST is a central reporting metric and is valid regardless of proportional hazards.

6. MARS achieves performance close to IPD meta-analysis while using only aggregate-level information.

7. The framework substantially broadens the scope of survival meta-analysis beyond conventional hazard-ratio pooling.

6. RESOLVE-IPD

Main Objective

The paper develops RESOLVE-IPD, a unified framework for reconstructing individual patient data (IPD) from published oncology trials and conducting uncertainty-aware subgroup meta-analysis.

The framework addresses two major limitations in existing reconstruction methods:

1. Reconstruction inaccuracies caused by digitization errors and unrealistic censoring assumptions.

2. Inability to reliably recover subgroup-level IPD when only aggregate subgroup statistics are reported.

To address these issues, RESOLVE-IPD introduces three major methodological components:

1. VEC-KM (Vector Kaplan–Meier Extraction)

2. CEN-KM (Censoring-Informed Kaplan–Meier Reconstruction)

3. MAPLE (Marginal Assignment of Plausible Subgroup Labels and Evidence Propagation)

Together these components form an end-to-end workflow for high-fidelity IPD reconstruction, subgroup label recovery, and uncertainty quantification, which allows for richer downstream analysis.

Motivation

IPD are critical for treatment effect estimation and followup studies like subgroup and meta-analysis, but are rarely publically available due to data ethics, so research must rely on published KM curves, risk tables, and subgroup summaries. The paper argues that existing reconstruction methods contain methodological weaknesses that can substantially bias downstream analyses.

Problems with Existing IPD Reconstruction Methods

The authors identify two broad classes of problems:

- Digitization Error from traditional workflows (human + floating point)

- Uniform Censoring Assumption: The dominant iKM algorithm assumes censoring is uniform within intervals, which is unrealistic:
  - Patients are often censored at scheduled visits
  - Administrative censoring occurs at specific times
  - Dropout patterns are not uniform

Additional Challenge: Subgroup Reconstruction

Many oncology publications report overall KM curves and subgroup statistics but not subgroup specific KM curves (in example with PD-L1 high vs low) creating a non-identifiable problem.

Multiple patient level subgroup assignments may reproduce the same HR or survival outcomes, so point identification is nullified. The paper emphasizes that this uncertainty must be quantified and propagated into downstream analyses.

Overview of RESOLVE-IPD

The framework consists of two major stages.

Stage 1

High-fidelity IPD reconstruction through VEC-KM and CEN-KM, outputting

$$(T_i, \delta_i, A_i)$$

where T_i = survival time, δ_i = event indicator, A_i = treatment group.

Stage 2

Uncertainty-aware subgroup recovery via MAPLE, outputting G_{MAPLE}, a set of plausible subgroup label assignments used for meta analysis, RMST/HR estimation, survival comparison.

VEC-KM

Purpose

Reduce digitization error by extracting information directly from vector graphics. Instead of raster images, VEC-KM uses vector paths embedded in PDFs or SVG files. Because vector graphics contain exact drawing instructions, extraction is independent of image resolution.

Advantages

Compared with pixel-based methods, there is no rounding artifacts, manual calibration. Higher coordinate precision (up to 6 decimnals) and censor mark detection as well.

Core Steps

1. KM Curve Extraction: Identify all line segments and select the longest connected paths as candidate curves
2. Axis Calibration: Locate x- and y- axes and construct maps between pixel space, time, and survival probability
3. Censor-Mark Detection: identify remaining short vector segments and classify them based on shape and proximity to KM curves

CEN-KM

Purpose

Reconstruct IPD while explicitly incorporating censoring information. (Replaces uniform censoring assumption)

Key Innovations

Explicit Censoring Times Rather than estimating censoring indirectly, CEN-KM uses censoring times extracted by VEC-KM.

Overlapping Censor Marks Clinical KM plots often contain overlapping censor symbols. A single visible mark may represent multiple censored patients, which existing methods tend to ignore.

Algorithmic Strategy

For each KM interval:

1. Count observed censor marks.
2. Generate candidate event counts.
3. Evaluate possible event-censor combinations.
4. Increment overlapping censors when necessary.
5. Select the configuration that best reproduces the published survival drop.

Risk-Table Alignment

When a reported number-at-risk value is encountered, reconstructed risk sets are compared with published values, censor counts are adjusted and exact risk-table consistency is enforced.

Advantages

Compared with iKM, this uses explicit censoring information and corrects floating point error, which produces more accurate curves.

MAPLE

Purpose

Marginal Assignment of Plausible subgroup Labels and Evidence Propogation (MAPLE) attempts to recover subgroup labels when subgroup KM curves are unavailable.

Key Idea

The paper explicitly rejects the idea of finding a single ”correct” subgroup labeling:

$$G_{\text{MAPLE}} = \{g : g \text{ satisfies published evidence}\}$$

where g denotes a patient-level subgroup assignment and multiple solutions are allowed.

Inputs

MAPLE uses reconstructed overall IPD and published information (subgroup sample sizes, hazard ratios, median survival, confidence intervals).

Optimization Objective

MAPLE searches for subgroup label assignments that reproduce τ^{tgt} where target quantities are hazard ratios, medium survival, and confidence intervals. Assignments minimizing reconstruction error are retained.

Refinement Using Low-Fidelity KM Curves

If subgroup KM curves exist only as raster images, approximate subgroup IPD is reconstructed and candidate labelings are filtered. The refined set is denoted G^*_{MAPLE}.

Interpretation

The output is not a single subgroup assignment. Instead:

$$G_{\text{MAPLE}} = \{g_1, g_2, \ldots, g_M\}$$

where every labeling is consistent with the available evidence. The resulting variability quantifies subgroup-reconstruction uncertainty.

Deterministic vs Optimization Pathways

The paper distinguishes two subgroup-recovery pathways:

- Deterministic: Used when subgroup KM curves are available, and produces most reliable subgroup IPD. Scenarios:
  - Target subgroup KM curve exists directly.
  - KM subtraction can recover the missing subgroup.

- Optimization: Used when only aggregate subgroup summaries are available, using MAPLE to produce an ensemble of plausible labelings

Esophageal Cancer Application

The framework is applied to advanced esophageal squamous cell carcinoma, with the focus on PD-L1 Low. Included trials are immunotherapy studies including KEYNOTE-181, ATTRACTION-3, and ESCORT. The goal is to reconstruct subgroup survival data, estimate treatment effects, and quantify uncertainty induced by reconstruction.

Main Findings

VEC-KM produced highly accurate coordinate extraction, and vector-based extraction significantly reduced digitization error.

CEN-KM improved recovery of censoring patterns, which produced curves that closely matched KM plots.

MAPLE successfully recovered multiple subgroup labelings and generated multiple data-compatible reconstructions.

Meta-Analysis

After uncertainty propagation, immunotherapy showed benefit relative to chemotherapy. Significant effects were particularly evident between 6 and 12 months.

Main Contributions

1. Introduces vector-based KM extraction through VEC-KM.
2. Eliminates the uniform-censoring assumption through CEN-KM.
3. Explicitly models overlapping censor marks.
4. Introduces MAPLE for subgroup-label reconstruction.
5. Treats subgroup reconstruction as a non-identifiable problem.
6. Produces a feasible set of subgroup assignments rather than a single solution.
7. Propagates reconstruction uncertainty into downstream meta-analysis.
8. Demonstrates the framework in a biomarker-defined oncology setting.

7. Transportable Inference Using Target Population Summary Statistics Under Covariate Shift

Main Objective

The paper studies transportability when:

- Individual-level data are available from a source population.
- Only summary statistics of covariates are available from a target population.

The goal is to perform valid statistical inference for the target population despite the absence of target individual-level data. The authors focus on estimating population quantities in the target population while accounting for:

1. Covariate shift between source and target populations.
2. Uncertainty in the reported target summary statistics.

A major contribution of the paper is that it explicitly incorporates uncertainty from target summary statistics into transportability procedures.

Motivation

Most existing transportability methods require both individual level source and target data. Example methods are inverse probability weighting or doubly robust transport estimators.

However, in many practical settings:

- Privacy regulations prohibit access to target individual-level data.
- Only aggregate demographic summaries are available.
- Registry data may provide means, proportions, and standard deviations rather than patient-level observations.

The paper addresses this setting directly.

Framework

Let:

- Y denote the outcome.
- X denote covariates in the source population.
- Y^* denote the outcome in the target population.
- X^* denote covariates in the target population.

The parameter of interest is $\mu^* = E(Y^*)$, the mean outcome in the target population.

Covariate Shift Assumption

The paper assumes covariate shift. The target distribution satisfies $f^*(y, x) = f(y, x)w(x)$ where:

- $f(y, x)$ is the source joint distribution.
- $f^*(y, x)$ is the target joint distribution.

- $w(x)$ is a density-ratio function.

Covariate distributions may differ, and conditional outcome models remain unchanged, therefore $f^*(y|x) = f(y|x)$. This is the transportability assumption in the paper.

Target Mean Representation

Under covariate shift:, $\mu^* = E\{w(X)Y\}$. This representation motivates weighting-based transport estimators. The problem therefore becomes estimating:

$$w(x) = \frac{f^*(x)}{f(x)}.$$

The difficulty is that source individual level covariates are observed but target individual level covariates are unavailable.

Available Information

The target population contributes:

$$\bar{\Phi}^* = \frac{1}{m}\sum_{j=1}^{m} \Phi(X_j^*)$$

where $\Phi(X)$ is a vector of known transformations of the covariates (usually moments or interaction summaries). The raw observations $X_1^*, \ldots, X_m^*$ are not observed.

Entropy Balancing Transportability

The first proposed approach extends entropy balancing to the transportability setting.

Core Idea

Construct weights w_i for source observations such that:

$$\sum_{i=1}^{n} w_i \Phi(x_i) = \bar{\Phi}^*.$$

so the weighted source moments exactly match target moments.

Optimization Problem

Entropy balancing solves:

$$\min \sum_{i=1}^{n} w_i \log(w_i)$$

subject to:

$$\sum_{i=1}^{n} w_i = 1 \quad \text{and} \quad \sum_{i=1}^{n} w_i \Phi(x_i) = \bar{\Phi}^*$$

Among all weight vectors satisfying moment constraints, choose the one closest to uniform weighting.

Resulting Weight Form

The solution takes the exponential-tilting form:

$$w_i = \frac{\exp\{\lambda^\top \Phi(x_i)\}}{\sum_{j=1}^n \exp\{\lambda^\top \Phi(x_j)\}}$$

for some Lagrange multiplier vector λ. This is one of the key mathematical results of the paper.

Transportability Estimator

After obtaining weights:

$$\hat{\mu}_{EB} = \sum_{i=1}^n w_i y_i.$$

This estimator transports the source outcome distribution to the target population.

Asymptotic Theory for Entropy Balancing

A major contribution is asymptotic inference. The paper proves:

$$\sqrt{n}(\hat{\mu}_{EB} - \mu^*)$$

is asymptotically normal. This permits confidence intervals, hypothesis testing, and variance estimation.

Variance Estimation Challenge

Existing transportability methods often assume target summaries are fixed. However $\bar{\Phi}^*$ is estimated from a finite target sample.

Therefore:

$$Var(\bar{\Phi}^*) \neq 0.$$

Ignoring this source of variability leads to underestimated standard errors and invalid confidence intervals, motivating the paper's variance correction methodology.

Variance Estimator

The authors derive an asymptotic variance estimator that accounts for source sampling variability, weight estimation variability, and target-summary uncertainty.

An important contribution is that the variance estimator only requires source individual-level data and target summary statistics. No target individual-level observations are needed.

Limitation of Entropy Balancing

Entropy balancing requires $\log\{w(x)\}$ to be representable as a linear combination of $\Phi(x)$.

Specifically:

$$\log\{w(x)\} = \lambda^\top \Phi(x).$$

This assumption may be restrictive.

Potential problems include nonlinear covariate shift, high-order interactions, and complex target distributions.

When these occur, moment matching may be insufficient and entropy balancing may be biased.

Flexible Covariate-Shift Modeling

To overcome these limitations, the paper develops a second method.

Instead of assuming

$$\log\{w(x)\} = \lambda^\top \Phi(x),$$

the density ratio is modeled as

$$w(x) = \pi(x; \alpha)$$

where α is an unknown parameter vector.

This permits much richer specifications, including nonlinear models, interaction models, and flexible semiparametric structures.

Augmented Objective Function

A key challenge is estimating α without observing target individual-level covariates.

The paper proposes an augmented optimization criterion. The criterion matches target moments and accounts for uncertainty in target summaries.

This augmentation is the key methodological innovation beyond entropy balancing.

Model Checking

The paper develops a specification test for $\pi(x; \alpha)$.

The resulting test statistic is shown to converge to a χ^2 distribution under the null hypothesis.

The purpose is to determine whether the chosen density-ratio model adequately captures covariate shift.

This provides a formal diagnostic unavailable in standard entropy balancing.

Relationship Between Methods

The paper shows that entropy balancing is a special case of the proposed framework.

Specifically,

$$\log\{\pi(x;\alpha)\} = \alpha^\top \Phi(x)$$

reduces the flexible method to entropy balancing.

Thus, entropy balancing is nested within the proposed framework, and the proposed method is strictly more general.

Efficiency Results

The paper establishes several theoretical comparisons.

When entropy balancing is correctly specified,

$$\log\{w(x)\} = \lambda^\top \Phi(x),$$

the proposed estimator is asymptotically at least as efficient as entropy balancing.

When entropy balancing is misspecified, entropy balancing may become inconsistent, while the proposed estimator remains consistent if $\pi(x;\alpha)$ is correctly specified.

This is one of the paper's main theoretical advantages.

Simulation Studies

The simulations evaluate bias, variance, coverage probability, and mean squared error.

Scenarios include correctly specified entropy balancing, misspecified entropy balancing, nonlinear covariate shift, and different levels of target-summary uncertainty.

Main Findings

When entropy balancing is correctly specified, both methods perform well and exhibit similar bias and coverage.

When entropy balancing is misspecified, entropy balancing becomes biased and coverage deteriorates.

The proposed method remains consistent, approximately unbiased, and properly calibrated.

The advantage becomes more pronounced as covariate shift becomes increasingly nonlinear.

SEER Breast Cancer Application

The methods are applied to the Surveillance, Epidemiology, and End Results (SEER) breast cancer data.

The purpose is to estimate target-population outcomes when only summary statistics are available from the target population.

The application demonstrates practical implementation, variance correction, and flexible transportability modeling.

The authors show that accounting for uncertainty in target summaries materially affects inference. Confidence intervals become wider after incorporating target-summary uncertainty.

Key Insights

A recurring theme throughout the paper is Target summary statistics should not be treated as fixed quantities.

Instead, $\bar{\Phi}^*$ must be viewed as an estimated quantity.

Ignoring its uncertainty can produce overconfident inference, underestimated variance, and poor coverage.

The proposed framework corrects this issue.

Limitations

The paper assumes covariate shift, correct specification of the density-ratio model, and availability of informative target summaries.

The methods do not address outcome shift, label shift, concept drift, or missing target summaries.

The framework also requires sufficient overlap between source and target populations.

Main Takeaways

1. The paper studies transportability when only target covariate summaries are available.
2. Entropy balancing naturally extends to this setting through moment matching.
3. The authors establish asymptotic normality and valid variance estimation for entropy balancing.
4. Target-summary uncertainty must be incorporated into inference.
5. A flexible density-ratio modeling framework generalizes entropy balancing.
6. Entropy balancing becomes a special case of the proposed method.
7. The proposed estimator remains valid under more general forms of covariate shift.
8. Simulations demonstrate improved robustness under nonlinear covariate shift.

9. The framework enables valid target-population inference without access to target individual-level data.

8. StatLLM: Evaluating LLMs for Statistical Analysis

Main Objective

StatLLM is an open-source benchmark dataset designed to evaluate the ability of Large Language Models (LLMs) to perform statistical programming and statistical analysis.

The paper addresses a major gap in the evaluation literature:

> There is no standardized benchmark for evaluating LLM performance in statistical programming languages such as SAS and R.

Most existing code benchmarks focus on Python, general software engineering tasks, and algorithmic programming problems.

StatLLM instead focuses on statistical reasoning, statistical programming, statistical methodology selection, and data analysis workflows.

The benchmark is intended to evaluate whether LLMs can correctly translate statistical questions into executable statistical analyses.

Motivation

Recent LLMs can generate code for Python, R, SAS, and SQL.

However, successful statistical analysis requires much more than code generation.

A model must:

1. Understand the statistical objective.
2. Identify the appropriate method.
3. Generate syntactically correct code.
4. Produce executable code.
5. Produce correct statistical output.

A model may generate code that appears reasonable but uses the wrong statistical method, violates assumptions, produces incorrect results, or fails to execute.

Therefore evaluating statistical AI systems requires domain-specific benchmarks.

StatLLM Dataset Structure

The benchmark contains three primary components.

Statistical Analysis Tasks

The dataset contains 207 statistical analysis tasks constructed from 65 real datasets. Each task includes a dataset description, problem description, and human-verified SAS solution.

LLM-Generated Code

The benchmark includes code generated by ChatGPT 3.5, ChatGPT 4.0, and Llama 3.1 70B. Each model receives a dataset description, variable information, and statistical objective. The resulting SAS code is stored and evaluated.

Human Evaluation Scores

Every generated solution is reviewed by statistics experts. Human evaluators assess code quality, executability, and output correctness. The resulting scores form the benchmark's ground truth.

Statistical Topics Covered

The benchmark spans a broad range of statistical analyses, ranging from data management, descriptive statistics, hypothesis testing, GLMs, and variable selection.

Advanced methods like survival analysis and Cox PH models, along with nonparametric methods, are covered.

Why Statistical Benchmarks are Different

Traditional code benchmarks primarily evaluate whether a system can correctly map an input to the desired output.

Statistical analysis introduces additional layers. A model must understand the question being asked, select an appropriate statistical method, generate correct code, and ultimately produce valid statistical inference.

As a result, an answer may be syntactically correct while still being statistically wrong.

For example, a model may use linear regression instead of logistic regression, use a t-test instead of ANOVA, or ignore censoring in survival data.

Therefore statistical evaluation requires domain-specific criteria.

Human Evaluation Framework

One of the major contributions of StatLLM is the use of expert human evaluation.

The benchmark does not rely solely on automated metrics. Instead, statisticians manually score generated code.

The paper identifies several evaluation dimensions.

Code Quality

Code quality measures statistical appropriateness, logical correctness, method selection, and code structure.

The central question is whether the model chose the correct statistical procedure.

Executability

Executability measures whether code runs successfully and whether syntax errors, missing arguments, or invalid procedures are present.

The central question is whether the code can actually be executed.

Output Accuracy

Output accuracy measures whether the produced output answers the question, whether the correct analysis was performed, and whether the reported results are correct.

The central question is whether the code produced the correct statistical result.

Overall Score

The benchmark combines multiple dimensions into a total evaluation score.

This provides a holistic assessment of statistical-analysis performance.

Evaluation Metrics for Statistical LLMs

The paper motivates several categories of evaluation metrics. These are useful beyond StatLLM and are directly relevant to future statistical AI systems.

Method Selection Accuracy

Arguably the most important metric.

Define

$$MSA = \frac{\text{Correct Method Selections}}{\text{Total Tasks}}.$$

Examples include selecting logistic regression when appropriate, choosing a Cox model for survival analysis, or using ANOVA for group comparisons.

A statistically incorrect method should be counted as failure even if the code executes.

Code Executability Rate

Define

$$ER = \frac{\text{Executable Programs}}{\text{Total Programs}}.$$

This metric measures syntax correctness and software compatibility.

It is analogous to pass rates in software-engineering benchmarks.

Output Accuracy Rate

Define

$$OAR = \frac{\text{Correct Outputs}}{\text{Total Tasks}}.$$

This evaluates whether the generated code ultimately solves the statistical problem.

Statistical Correctness

Statistical correctness measures whether assumptions are respected, proper estimands are reported, and statistical procedures are correctly specified.

This is more important than simple code correctness.

Human Evaluation Score

A general framework can be written as

$$Score = f(Quality, Executability, Accuracy).$$

Human evaluation remains the gold standard.

Agreement with Human Solutions

Suppose C_{LLM} is generated code and C_{Human} is verified code.

One can compare procedure selection, parameter choices, and output equivalence.

Importantly, exact code matching is not required. Different code can produce the same correct analysis.

Limitations of Traditional NLP Metrics

The paper highlights a major issue.

Many existing NLP benchmarks rely on text-similarity metrics such as BLEU and ROUGE.

BLEU (Bilingual Evaluation Understudy) measures overlap of n-grams between a generated response and a reference response. Higher BLEU scores indicate greater textual similarity.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) measures overlap between generated and reference text, often emphasizing recall of important words or phrases.

These metrics are useful for translation and summarization tasks, where textual similarity is closely related to correctness.

However, they may not be appropriate for statistical code.

Two SAS programs may be completely different textually but perform identical analyses. Conversely, two programs may appear very similar while producing different statistical conclusions.

Thus,

$$\text{Text Similarity} \neq \text{Statistical Correctness.}$$

This motivates specialized evaluation metrics for statistical programming.

Benchmarking Framework for Statistical AI

The paper implicitly suggests a hierarchy of evaluation:

1. Can the code run?
2. Is the method correct?
3. Is the statistical output correct?
4. Is the interpretation correct?

Future statistical AI benchmarks should ideally evaluate all four levels.

Connection to Statistical Copilots

The authors argue that benchmarks such as StatLLM are necessary for developing AI-assisted statistical software, statistical copilots, and natural-language data-analysis systems.

Future systems may allow users to specify analyses through natural language rather than programming directly.

Reliable benchmarks are required before such systems can be trusted.

Key Takeaways

1. StatLLM is one of the first benchmark datasets specifically designed for statistical programming evaluation.
2. The benchmark contains 207 statistical tasks spanning 65 datasets.
3. Tasks cover topics ranging from descriptive statistics to survival analysis.
4. Human experts evaluate generated code rather than relying solely on automated metrics.

5. Statistical evaluation requires more than syntax checking.
6. Method selection is often more important than code similarity.
7. Executability, correctness, and output accuracy should all be evaluated separately.
8. Traditional NLP metrics may be insufficient for statistical programming tasks.
9. Statistical AI systems should ultimately be evaluated on the quality of their statistical reasoning rather than text generation alone.

9. AUTOMAT: A Benchmark for Agentic Reproduction

Main Objective

AUTOMAT is a benchmark designed to evaluate whether LLM-based coding agents can reproduce computational materials science claims from scientific papers and associated artifacts. The key distinction is that AUTOMAT evaluates end-to-end scientific reproduction, workflow recovery, scientific validation, and long-horizon execution rather than simple code generation or tool use.

The central question is:

Can an autonomous agent reconstruct, execute, and verify a scientific claim from a published paper?

Motivation

Most existing AI benchmarks focus on programming problems like code generation, SWE tasks, and tool usage.

Examples include standard coding benchmarks and scientific coding benchmarks such as SciCode.

However, scientific reproducibility requires much more than writing code.

An agent must:

- Understand scientific claims and identify the workflow, software requirements and inputs
- Execute computational experiments and interpret results correctly
- Determine whether the claim is supported

AUTOMAT was developed specifically to evaluate these capabilities.

Core Benchmark Representation

Each benchmark instance is represented as $x = (q, p, m, a)$ where:

q = scientific claim, p = source paper, m = metadata, a = available artifacts (scripts, datasets, custom code, config, simulation outputs)

The agent does not receive expert reproduction instructions. Instead, it must independently determine how to reproduce the claim.

Computational Materials Science Domains

Claims span multiple subfields like density field theory (DFT), molecular dynamics, statistical learning methods, and discrete dislocation dynamics. These domains were selected because they represent diverse reproducibility challenges.

Claim Selection Philosophy

Claims are chosen by subject matter experts (SMEs). The focus is on numerical claims that support a paper's main scientific conclusions. Examples include formation energies, prediction accuracies, diffusion coefficients, and thermodynamic quantities. The benchmark intentionally emphasizes quantitative verification rather than qualitative interpretation.

Claim Types

AUTOMAT contains three claim categories.

From-Paper Reproduction

The agent receives a scientific paper and meta data, but the workflow has to be reconstructed from the publication itself and without artifacts.

This is the most difficult setting because workflow details are often underspecified.

From-Artifact Reproduction

The agent receives everything from From-Paper reproduction but gets essential artifacts like code repositories, trained models, and simulation inputs.

The challenge becomes execution and adaptation rather than workflow discovery.

From-Artifact Interpretation

The agent receives completed outputs. The agent must determine whether the provided results support the scientific claim.

This tests scientific reasoning rather than computational execution.

Verification Categories

AUTOMAT further categorizes claims by verification type.

Scalar Metric Match

Verify a numerical quantity (accuracy, energy, error rate)

Comparative or Ordering Claims

Verify relative comparisons.

Examples:

- Method A outperforms Method B
- Material X is more stable than Material Y

Range or Bound Verification

Verify whether a result lies within a reported range.

Structural or Qualitative Verification

Verify qualitative scientific behavior (response curve shape/stability patterns)

Figure Reproduction

Verify whether an agent can recreate a published figure.

Agent Execution Framework

Given a task $x = (q, p, m, a)$, an agent performs a reproduction attempt:

$$\tau = \text{Agent}_{rep}(x).$$

The trajectory τ contains commands executed, files generated, logs and intermediate outputs, and final reports. The benchmark therefore evaluates complete scientific workflows rather than final answers alone.

Task-Specific Orchestrated Agent

The authors build a specialized reproduction agent with the goal of improved auditability and workflow control.

The workflow contains four phases:

1. Read-only planning
2. Environment preparation
3. Deterministic execution with failure diagnosis
4. Result extraction and self-assessment

Importantly, this agent is not given additional scientific knowledge.

Evaluation Framework

A separate evaluator agent assesses reproduction quality. Unlike a static judge, the evaluator can read generated files and compare against expert solutions. The evaluation function is

$$y = \text{Agent}_{eval}(g, q, p, \tau),$$

where:

- g = SME-authored ground-truth reproduction procedure
- q = claim
- p = paper
- τ = reproduction trajectory

Overall Reproducibility Score

Each run receives a score from 1 to 5:

5	Fully reproduced
4	Mostly reproduced
3	Partially reproduced
2	Mostly failed
1	Failed

A successful reproduction is defined as: Score ≥ 4

Evaluation Dimensions

The overall score is supported by five dimensions:

- Methodological Fidelity: Did the agent follow an appropriate scientific procedure?
- Execution Competence: Could the workflow be executed successfully?
- Result Accuracy: Were the reproduced results correct?
- Completeness: Did the agent complete all required steps?
- Scientific Rigor: Did the agent provide sufficient evidence and scientific justification?

Judge Calibration

To validate the automated evaluator, the authors compare it against blinded SME evaluations.

A subset of n=40 runs is independently reviewed by experts and found Quadratic-weighted kappa: 0.69 and Within-1 score agreement: 0.80. This indicates substantial agreement between automated and human evaluation.

Research Questions

The benchmark investigates three questions:

1. How well can current coding agents reproduce scientific claims?
2. Does task-specific orchestration improve reproducibility performance?
3. What failure modes prevent successful reproduction?

Models Evaluated

Five systems were evaluated:

- Claude Code + Opus 4.6
- Claude Code + Sonnet 4.6
- Claude Code + Kimi K2.5
- Codex CLI + GPT-5.4
- Orchestrated Agent + Sonnet 4.6

The orchestrated agent uses the same underlying model as Claude Code Sonnet, enabling controlled comparisons.

Main Results

Overall performance remains limited.

Mean reproducibility scores:

Agent	Mean Score
Claude Code + Opus	3.52
Claude Code + Sonnet	3.12
Orchestrated + Sonnet	2.92
Claude Code + Kimi	2.75
Codex + GPT-5.4	2.44

The best system achieved a success rate of approximately 54.1%.

The weakest achieved approximately 23.5 %.

Thus, no system was reliably capable of autonomous scientific reproduction.

Performance by Claim Type

From-paper reproduction was the most difficult setting. Mean scores ranged roughly from 1.5 to 2.2

Success rates were near zero for most systems. The primary bottleneck is workflow recovery from incomplete paper descriptions.

From-artifact reproduction was substantially easier.

Mean scores ranged approximately 3.1-4.1 ith success rates of 39% to 77%. Providing executable artifacts removes much of the ambiguity.

From-artifact interpretation remained challenging despite available outputs.

Success rates ranged from 33% to 50%. This suggests that scientific interpretation remains difficult even when execution is unnecessary.

Effect of Orchestration

A central finding is that workflow orchestration did not substantially improve overall performance.

When comparing:

- Orchestrated Agent + Sonnet
- Claude Code + Sonnet

most benchmark instances resulted in ties.

The orchestrated system showed improvement only in scientific rigor.

The free-form Claude Code workflow often performed better on execution-related tasks.

The authors conclude that orchestration introduces trade-offs rather than universally improving performance.

Key Failure Modes

Several recurring challenges emerge:

- Workflow reconstruction from incomplete papers
- Missing software dependencies
- Long-horizon execution failures
- Incorrect scientific interpretation
- Inability to adapt workflows when execution errors occur

The benchmark demonstrates that scientific reproducibility requires substantially more than coding ability.

Key Takeaways

- AUTOMAT evaluates end-to-end scientific reproducibility rather than isolated coding tasks.
- Tasks require workflow reconstruction, execution, validation, and interpretation.

- Three claim types are considered: from-paper reproduction, from-artifact reproduction, and from-artifact interpretation.
- Reproducibility is evaluated using methodological fidelity, execution competence, result accuracy, completeness, and scientific rigor.
- Current coding agents succeed on only a minority of benchmark tasks.
- From-paper reproduction remains the dominant bottleneck.
- Providing executable artifacts substantially improves performance.
- Scientific interpretation remains difficult even when outputs are available.
- Task-specific orchestration improves rigor but does not substantially improve overall reproducibility.
- Scientific reproducibility remains a challenging frontier for autonomous AI systems.

Part II

Notes

10. Survival Analysis

Overview

Survival analysis studies the time until an event occurs. Common events include death, disease progression, relapse, equipment failure, and customer churn. Unlike standard regression, survival analysis must account for time-to-event outcomes, censoring, and changing risk over time.

The primary random variable is T, the event time.

Censoring

Survival data are often only partially observed.

For right censoring, we observe $Y = \min(T, C)$ with event indicator $\delta = I(T \leq C)$

If $\delta = 1$, the event is observed. If $\delta = 0$, the observation is censored.

Other forms include left censoring ($T < C$) and interval censoring ($L < T < R$).

A fundamental assumption is independent censoring $T \perp C$. Violations produce informative censoring and may bias inference.

Core Survival Quantities

The survival function is $S(t) = P(T > t)$, the probability of surviving beyond time t.

The cumulative distribution function is $F(t) = P(T \leq t) = 1 - S(t)$.

For continuous outcomes,

$$f(t) = -\frac{dS(t)}{dt}.$$

The hazard function is

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t},$$

which represents the instantaneous event risk among individuals still at risk.

The cumulative hazard is

$$H(t) = \int_0^t h(u)\, du.$$

These quantities are related through $S(t) = e^{-H(t)}$ and $f(t) = h(t)S(t)$.

Kaplan–Meier Estimation

The Kaplan–Meier estimator is the standard nonparametric estimator of the survival function:

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right),$$

where d_j is the number of events and n_j is the number at risk at time t_j. The estimator is nonparametric, handles right censoring, and produces a stepwise estimate of the survival curve. The median survival time is

$$t_{0.5} = \inf\{t : S(t) \leq 0.5\}.$$

Nelson–Aalen Estimation

The Nelson–Aalen estimator estimates cumulative hazard:

$$\hat{H}(t) = \sum_{t_j \leq t} \frac{d_j}{n_j}.$$

A useful approximation is $\hat{S}(t) \approx e^{-\hat{H}(t)}$.

Comparing Survival Curves

The log-rank test is the most common test for comparing survival curves. The null hypothesis is

$$H_0 : S_1(t) = S_2(t).$$

The test compares observed and expected events across groups and is most powerful when proportional hazards holds.

Parametric Survival Models

Parametric models assume a specific distribution for T:

The *exponential* model assumes a constant hazard $h(t) = \lambda$ with survival function $S(t) = e^{-\lambda t}$.

The *Weibull* model assumes

$$h(t) = \lambda \gamma t^{\gamma - 1}$$

When $\gamma = 1$, the model reduces to the exponential model. Values of $\gamma > 1$ imply increasing hazard, while $\gamma < 1$ implies decreasing hazard.

The log-normal model assumes

$$\log(T) \sim N(\mu, \sigma^2)$$

and allows non-monotone hazards.

Accelerated Failure Time Models

Accelerated Failure Time (AFT) models act directly on survival times:

$$\log(T) = X^\top \beta + \varepsilon.$$

Covariates accelerate or decelerate survival time rather than modifying hazards. A major advantage is that AFT models do not require proportional hazards.

Cox Proportional Hazards Model

The Cox model is the most widely used survival model:

$$h(t|X) = h_0(t) \exp(X^\top \beta).$$

Here $h_0(t)$ is the baseline hazard and $\exp(X^\top \beta)$ represents relative risk.

For two individuals,

$$HR = \frac{h(t|X_1)}{h(t|X_2)} = \exp[(X_1 - X_2)^\top \beta].$$

The hazard ratio measures relative instantaneous risk.

Proportional Hazards Assumption

The key Cox assumption is that hazard ratios remain constant over time:

$$HR(t) = HR \quad \text{or} \quad \frac{h_1(t)}{h_2(t)} = \text{constant}$$

Common signs of violation include crossing Kaplan–Meier curves, delayed treatment effects, and immunotherapy-type survival patterns. Common diagnostics include Schoenfeld residuals, log-minus-log plots, and time-varying coefficient models.

Time-Varying Covariates

The Cox model can be extended to time-varying covariates:

$$h(t) = h_0(t) \exp[X(t)^\top \beta].$$

Examples include biomarker measurements, medication usage, and blood pressure.

Restricted Mean Survival Time

Restricted Mean Survival Time (RMST) is a popular alternative to hazard ratios:

$$RMST(\tau) = \int_0^\tau S(t)\, dt.$$

RMST represents expected survival accumulated before time horizon τ.

A common treatment effect is

$$\Delta_{RMST} = RMST_T - RMST_C.$$

Unlike hazard ratios, RMST does not require proportional hazards and remains interpretable when survival curves cross.

Competing Risks

Competing risks occur when multiple event types are possible and the occurrence of one event prevents another. Standard Kaplan–Meier methods may overestimate incidence in this setting. Common approaches include cause-specific hazard models and Fine–Gray models.

Frailty Models

Frailty models introduce unobserved random effects:

$$h(t|X, Z) = Zh_0(t)\exp(X^\top \beta),$$

where Z represents latent heterogeneity. These models account for unobserved differences in risk across individuals.

Bayesian Survival Analysis

Bayesian survival models place priors on hazards, survival functions, or regression coefficients.

Examples include Bayesian Cox models, piecewise exponential models, NTR processes, and Dirichlet process mixtures.

These approaches provide full uncertainty quantification and highly flexible hazard modeling.

Key Assumptions

The most important assumptions in survival analysis are independent censoring, correct model specification, proportional hazards (for Cox models), independent observations, and positivity/overlap in causal survival settings.

Key Takeaways

1. Survival analysis studies time-to-event outcomes.
2. Censoring is the defining challenge.
3. The survival function $S(t)$ gives survival probabilities.
4. The hazard function $h(t)$ measures instantaneous risk.
5. Kaplan–Meier estimation is the standard nonparametric survival estimator.
6. Cox regression is the dominant semiparametric model.

7. Hazard ratios require proportional hazards.
8. RMST provides a robust alternative when PH fails.
9. Frailty models account for unobserved heterogeneity.
10. Bayesian survival models provide flexible uncertainty-aware inference.

11. Meta-Analysis Review

What is Meta-Analysis?

Meta-analysis is a statistical framework for combining evidence from multiple studies that investigate the same scientific question.

Rather than relying on a single study, meta-analysis synthesizes information across studies to obtain more precise effect estimates, greater statistical power, improved generalizability, and better understanding of heterogeneity.

Conceptually, multiple studies are combined into one overall estimate.

The fundamental idea is that each study provides partial information about an underlying treatment effect.

Why Meta-Analysis Exists

Individual studies often suffer from small sample sizes, limited power, conflicting conclusions, and population-specific results. Meta-analysis pools information across studies to reduce uncertainty. For example:

Study	Effect Estimate	Standard Error
A	0.20	0.15
B	0.35	0.18
C	0.10	0.12

Each study alone is noisy. Meta-analysis combines them into a single estimate with smaller uncertainty.

Basic Setup

Suppose there are K studies. For study k, $\hat{\theta}_k$ denotes the estimated treatment effect. Examples include mean differences, odds ratios, risk ratios, hazard ratios, and RMST differences. Associated uncertainty is represented by s_k^2, where s_k is the standard error of $\hat{\theta}_k$. The goal is to estimate θ using all studies simultaneously.

Common Effect Measures

Mean Difference

For continuous outcomes (blood pressure reduction, weight loss, and test scores): $MD = \bar{Y}_T - \bar{Y}_C$.

Standardized Mean Difference

The standardized mean difference is used when studies measure outcomes on different scales:

$$SMD = \frac{\bar{Y}_T - \bar{Y}_C}{s_{pooled}}.$$

Common interpretations are 0.2 as a small effect, 0.5 as a moderate effect, and 0.8 as a large effect.

Risk Ratio

For binary outcomes:

$$RR = \frac{P(Y = 1|T)}{P(Y = 1|C)}.$$

Here $RR = 1$ indicates no effect, $RR < 1$ indicates treatment reduces risk, and $RR > 1$ indicates treatment increases risk.

Odds Ratio

The odds ratio is

$$OR = \frac{P(Y = 1|T)/(1 - P(Y = 1|T))}{P(Y = 1|C)/(1 - P(Y = 1|C))}.$$

It is often used in case-control studies and logistic regression.

Hazard Ratio

For survival outcomes:

$$HR = \frac{h_T(t)}{h_C(t)}.$$

Here $HR = 1$ indicates identical hazards, $HR < 1$ indicates treatment benefit, and $HR > 1$ indicates harm. Hazard ratios assume proportional hazards when interpreted globally.

The Core Meta-Analysis Estimator

The most basic estimator is a weighted average:

$$\hat{\theta}_{MA} = \sum_{k=1}^{K} w_k \hat{\theta}_k.$$

The weights satisfy $\sum_{k=1}^{K} w_k = 1$.

The entire field of meta-analysis largely revolves around how these weights are chosen.

Why Weight Studies?

Not all studies provide equal information. For example:

Study	Sample Size	SE
A	100	0.20
B	1000	0.05

Study B is much more precise, so $w_B > w_A$.

Meta-analysis assigns more weight to studies with smaller uncertainty.

Inverse Variance Weighting

The most common weighting scheme is

$$w_k = \frac{1/s_k^2}{\sum_{j=1}^{K} 1/s_j^2}.$$

Small variance implies large weight, while large variance implies small weight.

The pooled estimator becomes

$$\hat{\theta}_{MA} = \frac{\sum_{k=1}^{K} \hat{\theta}_k/s_k^2}{\sum_{k=1}^{K} 1/s_k^2}.$$

Variance of the Pooled Estimate

The variance of the pooled estimate is

$$Var(\hat{\theta}_{MA}) = \frac{1}{\sum_{k=1}^{K} 1/s_k^2}.$$

The standard error is

$$SE(\hat{\theta}_{MA}) = \sqrt{\frac{1}{\sum_{k=1}^{K} 1/s_k^2}}.$$

Confidence intervals follow

$$\hat{\theta}_{MA} \pm 1.96 \times SE(\hat{\theta}_{MA}).$$

What is Heterogeneity?

A central question in meta-analysis is whether all studies are estimating the same effect. If all studies estimate the same treatment effect, then $\theta_1 = \theta_2 = \cdots = \theta_K$, and there is little heterogeneity. If effects vary, then $\theta_1 \neq \theta_2 \neq \cdots \neq \theta_K$, and there is heterogeneity.

Sources of Heterogeneity

Differences may arise from patient populations, inclusion criteria, treatment protocols, follow-up durations, biomarker prevalence, and study quality. Heterogeneity is often the most important issue in practice.

Cochran's Q Statistic

One measure of heterogeneity is

$$Q = \sum_{k=1}^{K} w_k(\hat{\theta}_k - \hat{\theta}_{MA})^2.$$

Large values indicate disagreement across studies, while small values indicate consistency.

Under homogeneity:

$$Q \sim \chi^2_{K-1}.$$

I-Squared

A more interpretable measure is

$$I^2 = \max\left(0, \frac{Q - (K-1)}{Q}\right) \times 100\%.$$

Common interpretations are 0–25% as low heterogeneity, 25–50% as moderate heterogeneity, 50–75% as substantial heterogeneity, and 75%+ as high heterogeneity.

Forest Plots

The standard visualization is the forest plot. Each study contributes a point estimate and confidence interval. The pooled estimate is displayed at the bottom. Horizontal lines represent uncertainty, squares represent study estimates, and the diamond represents the pooled estimate.

Publication Bias

Meta-analysis assumes observed studies represent all completed studies. In practice, positive studies are more likely to be published, while negative studies are more likely to remain unpublished. This creates publication bias.

Funnel Plots

Publication bias is commonly assessed using a funnel plot. The plot places $\hat{\theta}_k$ against $SE(\hat{\theta}_k)$. Without bias, the plot appears symmetric. With bias, the plot becomes asymmetric.

Aggregate Data vs Individual Participant Data

Meta-analysis can be conducted using aggregate data or individual participant data.

Aggregate Data (AD)

Aggregate-data meta-analysis uses published summaries only, such as hazard ratios, odds ratios, and means. It is easy to obtain and computationally simple, but suffers from information loss and limited subgroup analysis.

Individual Participant Data (IPD)

IPD meta-analysis uses original patient-level observations. It is more flexible, more powerful, and allows treatment-effect heterogeneity analysis, but is difficult to obtain and resource intensive. IPD meta-analysis is often considered the gold standard.

Meta-Analysis for Survival Data

Historically, survival meta-analysis relies on $\log(HR)$ and corresponding standard errors.

The pooled hazard-ratio estimator is

$$\hat{\beta} = \frac{\sum_k w_k \hat{\beta}_k}{\sum_k w_k},$$

where $\hat{\beta}_k = \log(HR_k)$.

Modern approaches increasingly use reconstructed survival curves, RMST, time-varying treatment effects, and Bayesian survival models. These methods relax proportional hazards assumptions.

12. Meta-Analysis Models

Overview

The central goal of meta-analysis is to combine study-specific estimates $\hat{\theta}_1, \ldots, \hat{\theta}_K$ into a single summary estimate. Different meta-analysis models differ primarily in how they treat the underlying true study effects $\theta_1, \ldots, \theta_K$. The key question is whether all studies estimate the same treatment effect or whether the treatment effect varies across studies. This distinction leads to fixed-effect models, random-effects models, Bayesian meta-analysis, meta-regression, IPD meta-analysis frameworks, and survival-specific meta-analysis methods.

Fixed-Effect Models

Core Assumption

A fixed-effect model assumes that every study estimates the same true effect:

$$\theta_1 = \theta_2 = \cdots = \theta_K = \theta.$$

Thus,

$$\hat{\theta}_k = \theta + \varepsilon_k, \qquad \varepsilon_k \sim N(0, s_k^2).$$

All variability between studies is assumed to arise from sampling error. There is no between-study heterogeneity.

Interpretation

The fixed-effect estimator answers: what is the common treatment effect shared by all studies? This interpretation is only valid if all studies truly estimate the same effect.

Estimator

Using inverse-variance weighting,

$$w_k = \frac{1/s_k^2}{\sum_{j=1}^K 1/s_j^2},$$

the pooled estimate is

$$\hat{\theta}_{FE} = \sum_{k=1}^K w_k \hat{\theta}_k.$$

Equivalently,

$$\hat{\theta}_{FE} = \frac{\sum_{k=1}^K \hat{\theta}_k / s_k^2}{\sum_{k=1}^K 1/s_k^2}.$$

The variance is

$$Var(\hat{\theta}_{FE}) = \frac{1}{\sum_{k=1}^K 1/s_k^2}.$$

The model is simple, computationally efficient, and highly precise when homogeneity holds. However, it is often unrealistic, ignores heterogeneity, and can produce overconfident intervals.

Random-Effects Models

Motivation

Clinical studies often differ in patient populations, inclusion criteria, follow-up length, and treatment implementation. Therefore, $\theta_1, \ldots, \theta_K$ may not be identical. Random-effects models explicitly allow study-specific treatment effects.

Hierarchical Model

The random-effects model has two levels:

$$\hat{\theta}_k | \theta_k \sim N(\theta_k, s_k^2)$$

and

$$\theta_k \sim N(\mu, \tau^2).$$

Here μ is the overall average treatment effect and τ^2 is the between-study variance.

The model can be visualized as $\mu \to \theta_k \to \hat{\theta}_k$.

Interpretation

The pooled estimate answers: what is the average treatment effect across a population of studies?

Random-Effects Weights

Weights become

$$w_k = \frac{1}{s_k^2 + \tau^2}.$$

Compared with fixed-effect models, large studies receive less dominance and small studies receive more weight.

Estimating Heterogeneity

The central quantity is τ^2. Common estimators include DerSimonian-Laird, REML (Restricted Maximum Likelihood), Paule-Mandel, and Bayesian estimation.

Prediction Intervals

An important advantage of random-effects models is prediction. A future study effect is modeled as

$$\theta_{new} \sim N(\mu, \tau^2).$$

Confidence intervals describe uncertainty about μ, while prediction intervals describe uncertainty about θ_{new}.

Bayesian Meta-Analysis

Motivation

Frequentist random-effects models estimate μ and τ^2. Bayesian models treat these quantities as random variables.

Hierarchical Bayesian Model

The likelihood is

$$\hat{\theta}_k|\theta_k \sim N(\theta_k, s_k^2).$$

The study effects satisfy

$$\theta_k|\mu, \tau^2 \sim N(\mu, \tau^2).$$

Priors are placed on the hyperparameters:

$$\mu \sim p(\mu), \qquad \tau \sim p(\tau).$$

The posterior is

$$p(\mu, \tau, \theta_1, \ldots, \theta_K|Data).$$

Bayesian meta-analysis provides full uncertainty quantification, natural hierarchical modeling, easy incorporation of prior information, and flexible extensions.

Posterior Inference

Posterior means such as $E(\mu|Data)$ often serve as pooled estimates. Posterior credible intervals replace confidence intervals. For example,

$$P(\mu \in [a, b]|Data) = 0.95.$$

Shrinkage

A key feature is partial pooling. Extreme study estimates are shrunk toward μ. The amount of shrinkage depends on study variance and heterogeneity.

One-Stage vs Two-Stage Meta-Analysis

Two-Stage Meta-Analysis

Two-stage meta-analysis is historically the most common framework.

In stage 1, treatment effects are estimated separately within each study, giving $\hat{\theta}_1, \ldots, \hat{\theta}_K$.

In stage 2, the study-specific estimates are pooled using a meta-analysis model.

The workflow is study analysis followed by meta-analysis.

This approach is easy to implement, computationally simple, and supported by standard software. However, it can lose information, make interaction modeling difficult, and be less efficient.

One-Stage Meta-Analysis

One-stage meta-analysis analyzes all participant data simultaneously. A general form is

$$Y_{ij} \sim f(\eta_{ij}) = f(\alpha_j + \beta_j T_{ij} + \gamma_j X_{ij})$$

where i indexes patients and j indexes studies. Random effects model study variation. One-stage models are often more efficient, better for interaction modeling, better for missing data, and more flexible.

Why One-Stage Matters

Many treatment-effect heterogeneity analyses require treatment-by-covariate interactions. These are easier to estimate within a one-stage framework.

Meta-Regression

Motivation

Suppose studies differ systematically by mean age, biomarker prevalence, treatment duration, or study quality. Meta-regression attempts to explain heterogeneity using study-level covariates.

Model

A random-effects meta-regression can be written as

$$\hat{\theta}_k = \beta_0 + \beta_1 z_k + u_k + \varepsilon_k,$$

where z_k is a study-level covariate and $u_k \sim N(0, \tau^2)$. The coefficient β_1 represents how the treatment effect changes as the study-level characteristic changes.

Ecological Bias

A major limitation is ecological bias. Study-level relationships need not equal patient-level relationships:

$$E(Y|X, \text{patient}) \neq E(Y|\bar{X}, \text{study}).$$

This is one major motivation for IPD meta-analysis.

Survival Meta-Analysis

Traditional Approach

Historically, survival meta-analysis pools $\log(HR_k)$. Let $\hat{\beta}_k = \log(HR_k)$. Then

$$\hat{\beta}_{MA} = \frac{\sum_k w_k \hat{\beta}_k}{\sum_k w_k}.$$

Limitation

Hazard ratios assume $HR(t) = HR$, which is the proportional hazards assumption. Violations occur when curves cross, delayed treatment effects exist, or immunotherapy effects emerge later.

Modern Survival Meta-Analysis

Modern survival meta-analysis increasingly uses reconstructed IPD, Kaplan–Meier curves, RMST, and time-varying hazards. Examples include MARS, Bayesian survival meta-analysis, and piecewise exponential models.

Restricted Mean Survival Time

RMST is defined as

$$RMST(\tau) = \int_0^\tau S(t)\, dt.$$

It represents expected survival accumulated before time horizon τ. The treatment effect is

$$\Delta_{RMST} = RMST_T - RMST_C.$$

RMST requires no proportional hazards assumption, is clinically interpretable, and is robust to crossing survival curves.

13. Extremal Inference

Overview

Extremal inference is a framework for statistical inference when the parameter of interest is not uniquely identified. Rather than estimating a single value θ, the available data and assumptions only imply that the parameter lies within a feasible set Θ.

The goal becomes estimating $\Theta = [\theta_L, \theta_U]$ or, more generally,

$$\Theta = \{\theta : \theta \text{ is compatible with the observed data}\}.$$

Instead of asking what the treatment effect is, extremal inference asks what the smallest and largest treatment effects are that remain consistent with the evidence. This framework appears throughout partial identification, missing-data problems, causal inference, sensitivity analysis, robust optimization, distributionally robust inference, and reconstruction uncertainty problems.

Motivation

Classical statistical inference assumes that enough information exists to identify θ. For example, one might estimate $\hat{\theta} = 0.35$. However, many real-world problems are underdetermined. Multiple underlying data-generating processes may explain the same observations. Examples include missing covariates, missing outcomes, aggregate-data reconstruction, unobserved confounding, and subgroup assignment uncertainty. In these situations, the data do not imply a unique parameter. Instead, the data imply a set of plausible parameters.

Identification vs Partial Identification

Point Identification

A parameter is point identified if $\Theta = \{\theta\}$. Only one value is compatible with the observed data. Examples include linear regression coefficients, means of fully observed variables, and standard treatment effects in randomized trials.

Partial Identification

A parameter is partially identified if $|\Theta| > 1$. Many values satisfy the available evidence. The data only determine bounds:

$$\theta_L \leq \theta \leq \theta_U.$$

The focus shifts from estimation of a point to estimation of a region.

Identification Regions

Suppose the parameter of interest is θ. The identification region is

$$\Theta = \{\theta : \theta \text{ is consistent with all observed constraints}\}.$$

For one-dimensional parameters, $\Theta = [\theta_L, \theta_U]$, where

$$\theta_L = \inf_{\theta \in \Theta} \theta, \qquad \theta_U = \sup_{\theta \in \Theta} \theta.$$

These are called the extremal values.

Basic Example

Suppose $Y \in [0,1]$ and 20% of observations are missing. Without additional assumptions, $0 \leq E(Y) \leq 1$. If the observed mean among observed subjects is 0.6, then the sharp bounds become

$$0.8(0.6) + 0.2(0) = 0.48 \text{ and } 0.8(0.6) + 0.2(1) = 0.68$$

The parameter is not identified, but the feasible region is.

Sharp vs Non-Sharp Bounds

Sharp Bounds

Bounds are sharp if every point inside the interval can be attained by some data-generating process consistent with the assumptions. Formally, $\Theta = [\theta_L, \theta_U]$, and every value inside the interval is feasible. Sharp bounds contain exactly the information provided by the assumptions.

Non-Sharp Bounds

Non-sharp bounds are wider than necessary and contain impossible values. Researchers generally seek sharp bounds whenever possible.

Optimization View

Many extremal inference problems can be written as optimization problems. Suppose $\theta = g(X)$ subject to constraints $X \in \mathcal{C}$. Then

$$\theta_L = \min_{X \in \mathcal{C}} g(X), \qquad \theta_U = \max_{X \in \mathcal{C}} g(X).$$

This perspective is especially important in reconstruction problems.

Connection to Causal Inference

Many causal parameters are only partially identified. Suppose $Y(1)$ and $Y(0)$ are potential outcomes. The average treatment effect is

$$ATE = E[Y(1) - Y(0)].$$

Without randomization or strong assumptions, ATE may not be point identified. Instead, $ATE \in [\tau_L, \tau_U]$. The resulting interval represents all causal effects compatible with the observed data.

Sensitivity Analysis vs Extremal Inference

Sensitivity analysis studies how conclusions change under violations of assumptions. Extremal inference studies all possible values under a fixed set of assumptions. In sensitivity analysis, assumptions vary. In extremal inference, the parameter varies within the feasible set implied by fixed assumptions.

Extremal Inference in Meta-Analysis

Meta-analysis typically assumes each study provides a unique effect estimate. However, reconstruction problems introduce additional uncertainty. Suppose g denotes a reconstructed dataset. Then treatment effects become $\hat{\tau}(g)$. If multiple reconstructions $g_1, \ldots, g_M$ are feasible, then the effect estimate is no longer unique. Instead,

$$\tau \in \{\hat{\tau}(g_1), \ldots, \hat{\tau}(g_M)\}.$$

Extremal inference summarizes this uncertainty using

$$\tau_L = \min_g \hat{\tau}(g), \qquad \tau_U = \max_g \hat{\tau}(g).$$

Connection to MAPLE

In MAPLE, $G = \{g_1, \ldots, g_M\}$ represents a collection of subgroup label assignments consistent with published evidence. Traditional analyses choose one labeling. Extremal inference instead studies

$$\min_{g \in G} \hat{\tau}(g) \qquad \text{and} \qquad \max_{g \in G} \hat{\tau}(g).$$

These represent the least favorable and most favorable reconstructions with respect to the treatment effect. This is directly connected to current extensions of RESOLVE-IPD.

Monte Carlo Approximation

In practice, G may be extremely large, so enumerating every reconstruction is impossible. A common approximation is to sample $g_1, \ldots, g_B \sim G$ and compute $\hat{\tau}(g_1), \ldots, \hat{\tau}(g_B)$.

Then approximate

$$\tau_L \approx \min_b \hat{\tau}(g_b), \qquad \tau_U \approx \max_b \hat{\tau}(g_b).$$

The issue is that the true extremum may never be sampled. This becomes increasingly problematic as dimensionality grows.

Worst-Case and Best-Case Analysis

Extremal inference is often described as best-case and worst-case analysis. The best case is $\max \hat{\tau}$, and the worst case is $\min \hat{\tau}$. These provide robustness guarantees. If treatment remains beneficial even under the worst-case reconstruction, conclusions become more credible.

Robust Optimization Connection

Extremal inference has close connections to robust optimization.

Classical estimation can be written as

$$\hat{\theta} = \arg\min L(\theta).$$

Robust optimization instead considers

$$\hat{\theta} = \arg\min_{\theta} \sup_{u \in \mathcal{U}} L(\theta, u),$$

where $\mathcal{U}$ represents uncertainty. Similarly, extremal inference studies $\sup_{\theta \in \Theta} \theta$ and $\inf_{\theta \in \Theta} \theta$.

Extremal Inference for Survival Analysis

Suppose $\tau(g)$ represents a survival estimand under reconstruction g. Possible estimands include hazard ratios, survival probabilities, RMST differences, and median survival differences.

The extremal problem becomes

$$\min_{g \in G} \tau(g) \qquad \text{and} \qquad \max_{g \in G} \tau(g).$$

This identifies the full range of treatment effects compatible with the reconstruction constraints.

Key Takeaways

Extremal inference is used when parameters are only partially identified. The goal is to characterize feasible regions rather than estimate a single value.

Identification regions are defined by all values consistent with observed data and assumptions, and extremal values are obtained through constrained optimization.

Sharp bounds contain exactly the information implied by the assumptions.

Many reconstruction problems naturally lead to partial identification. MAPLE already produces a feasible set of subgroup assignments, making extremal inference a natural extension.

Monte Carlo sampling may fail to identify true extrema in high-dimensional reconstruction spaces.

Worst-case and best-case analyses provide robustness guarantees, and extremal inference is likely to play a central role in future extensions of uncertainty-aware IPD reconstruction.

Part III

Reporting

14. 6/7: Initial Testing for EO

We start testing single-covariate extremal RMST, using first fully synthetic data. After getting the results from the dummy experiment, we can then move to glueing RESOLVE/MAPLE inputs. This first experiment:

1. simulates reconstructed IPD
2. assigns binary subrgoup labels Z
3. computes the RMST treatment effect
4. samples many feasible assignments
5. compares MC vs Annealing vs EO ranges

14.1. DGP Specification

The initial experiment uses a fully synthetic dataset designed to mimic a reconstructed clinical-trial IPD setting.

We generate $n = 200$ patients with approximately equal treatment allocation, $P(A_i = 1) = 0.5$, and a binary biomarker subgroup $Z_i \in \{0, 1\}$. Biomarker prevalence is fixed at $P(Z_i = 1) = 0.40$. For each patient, an event time and censoring time are generated from subgroup-specific survival distributions. The observed data consist of (T_i, δ_i, A_i), where T_i is the observed follow-up time, δ_i is the event indicator, and A_i is treatment assignment.

The true subgroup labels are retained only for evaluation purposes and are assumed unavailable to all reconstruction procedures. The reconstruction problem therefore consists of recovering information about the unknown label vector $Z = (Z_1, \ldots, Z_n)$ using only aggregate subgroup information.

14.2. RMST Definition

For any feasible subgroup assignment Z, treatment effect is defined using the restricted mean survival time (RMST) difference. Throughout the experiment, the truncation horizon is fixed at $\tau = 12$.

The treatment effect associated with assignment Z is the difference between treatment and null effects. RMST was chosen because it is clinically interpretable and is increasingly used in modern oncology trials. Numerical integration of the KM estimate is used to compute RMST values.

14.3. Feasible Assignment Generation

If the published study reports K biomarker-positive patients, then every feasible assignment must satisfy $\sum_i Z_i = K$. Later experiments incorporate arm-specific constraints,

$$\sum_{i:A_i=1} Z_i = K_1, \qquad \sum_{i:A_i=0} Z_i = K_0.$$

A feasible assignment is therefore any binary label vector satisfying all reported subgroup counts. At this stage no survival-summary constraints are imposed. Feasibility is therefore purely combinatorial and reflects only the published subgroup prevalence information.

14.4. Monte Carlo Baseline

A total of 10,000 feasible assignments are generated by randomly allocating subgroup labels while preserving the count constraints.

For each assignment:

1. Generate a feasible subgroup label vector Z.
2. Compute $\Delta_Z(\tau)$.
3. Store the resulting RMST treatment effect.

The empirical distribution is summarized using a five number summary and mean. This provides a baseline approximation to the treatment-effect distribution induced by reconstruction uncertainty.

14.5. Simulated Tempering

The algorithm begins with a feasible subgroup assignment and repeatedly proposes local modifications while preserving all count constraints. Each iteration:

1. Select one subgroup-positive patient and one subgroup-negative patient.
2. Swap their labels.
3. Compute the resulting treatment effect.
4. Accept improvements automatically.
5. Accept worsening moves with temperature-dependent probability.

The acceptance probability is $P(\text{accept}) = \exp(\delta/T)$, where $\delta = \Delta_{Z'}(\tau) - \Delta_Z(\tau)$ for maximization. The temperature is gradually reduced throughout the search, allowing early exploration and later exploitation. Two independent searches are performed:

- maximize $\Delta_Z(\tau)$,
- minimize $\Delta_Z(\tau)$.

The largest and smallest observed treatment effects provide approximations to the boundaries of the feasible treatment-effect region.

14.6. Extremal Optimization

Extremal Optimization (EO) is a boundary-seeking heuristic designed for combinatorial optimization problems. Unlike simulated tempering, EO does not rely on a temperature schedule. Instead, it repeatedly modifies poorly performing components of the current solution. The original EO implementation attempted to evaluate patient influence directly through repeated RMST recalculation. This approach was computationally prohibitive, requiring hundreds of additional Kaplan–Meier and RMST evaluations per iteration.

To make the search computationally feasible, a dummy surrogate fitness function was introduced. For maximization, patients are scored according to long survival times within the treatment arm and short survival times within the control arm. For minimization, the direction is reversed. After computing patient-level fitness values:

1. Patients are ranked from worst to best fitness.
2. Low-fitness patients are preferentially selected.
3. A feasible swap partner is chosen.
4. Labels are exchanged.
5. The exact RMST treatment effect is recomputed.

Importantly, the surrogate score is used only to guide the search. All reported treatment effects remain based on the exact RMST estimator. Consequently, EO should be interpreted as a heuristic optimization procedure rather than a guaranteed global optimization algorithm.

14.7. Comparison Summary

The true subgroup assignment produced an RMST treatment effect of 1.290. Monte Carlo sampling generated a treatment-effect distribution centered near the true value, with mean 1.060 and median 1.059. The observed Monte Carlo range was $[-1.839, 3.664]$. Simulated tempering substantially expanded the feasible range to $[-6.761, 6.774]$. Extremal Optimization produced the largest observed range, $[-7.941, 9.109]$. Distributional summaries were:

Method	Mean	Q05	Median	Q95
Monte Carlo	1.060	-0.239	1.059	2.295
Simulated Tempering	0.863	-4.390	1.064	5.506
Extremal Optimization	0.733	-7.323	0.588	8.669

All three methods produced similar central tendencies, suggesting that they are exploring broadly the same feasible assignment region. The optimization-based procedures identified substantially heavier tails than random feasible sampling. The extremal treatment effects identified by simulated tempering and EO lie far outside the region typically visited by Monte Carlo sampling.

These results suggest that random feasible sampling may adequately characterize the center of the reconstruction-induced treatment-effect distribution, but may substantially underestimate the full identification region $[\Delta_L, \Delta_U]$. The experiment therefore supports the extremal-inference perspective, where uncertainty is characterized not only through average treatment effects but also through the boundaries of the feasible treatment-effect region implied by subgroup reconstruction uncertainty.

1. Post-MAPLE Extremal Reconstruction

The goal is to find the most extreme subgroup assignments that still satisfy the same published constraints. In other words, given a feasible set $\mathcal{Z}$ already defined by MAPLE-style constraints, search for the minimum and maximum treatment effects still compatible with the evidence. The target quantities are $\Delta_L(\tau) = \min_{Z \in \mathcal{Z}} \Delta_Z(\tau)$ and $\Delta_U(\tau) = \max_{Z \in \mathcal{Z}} \Delta_Z(\tau)$, where $\Delta_Z(\tau)$ is the RMST treatment effect under assignment Z.

Core Idea Use MAPLE assignments as starting points for boundary search. The boundary search is constrained so that every proposed assignment must still satisfy the published evidence.

Algorithm Sketch

1. Run MAPLE or MAPLE-style Monte Carlo reconstruction to obtain feasible assignments $Z^{(1)}, \ldots, Z^{(B)}$.
2. Compute $\Delta_Z(\tau)$ for all sampled assignments.
3. Identify lower-tail and upper-tail assignments from the Monte Carlo sample.
4. Initialize extremal search from those tail assignments.
5. Propose label swaps that preserve subgroup counts and arm-specific counts.
6. Reject any assignment violating published subgroup survival constraints.
7. For maximization, keep moves that increase $\Delta_Z(\tau)$; for minimization, keep moves that decrease $\Delta_Z(\tau)$.
8. Continue until extrema stabilize across repeated starting points.
9. Report the MAPLE Monte Carlo range, optimized feasible range, and the gap between them.

Plausibility Layer The same framework can be run under increasingly strict constraint sets:

- $\mathcal{Z}_{\text{count}}$: assignments satisfying only subgroup counts.
- $\mathcal{Z}_{\text{summary}}$: assignments satisfying counts and published subgroup survival summaries.
- $\mathcal{Z}_{\text{plausible}}$: assignments satisfying counts, survival summaries, and clinical plausibility rules.

This gives nested bounds: $[\Delta_L^{count}, \Delta_U^{count}]$, $[\Delta_L^{summary}, \Delta_U^{summary}]$, and $[\Delta_L^{plausible}, \Delta_U^{plausible}]$.

2. Agentic Reconstruction Benchmark for Binary Subgroups

This extension treats subgroup reconstruction as an AI-agent benchmark. The agent receives reconstructed survival data and published subgroup summaries, then must reconstruct labels, characterize uncertainty, and identify edge cases. The benchmark is designed to test whether an AI system can behave like a statistical reconstruction assistant, not merely produce code.

Agent Input Each task gives the agent:

- reconstructed survival data (T_i, δ_i, A_i),
- published subgroup counts,
- treatment-arm subgroup counts,
- optional subgroup survival summaries,
- target estimand, such as RMST difference,
- hidden true labels for evaluation only.

Agent Output The agent must return:

- reconstructed labels $\hat{Z}$,
- a sample of feasible assignments,
- estimated central treatment effect,
- estimated lower and upper feasible bounds,
- code used to generate the reconstruction,
- explanation of constraints and assumptions.

First Metrics to Test Initial evaluation should focus on simple metrics:

- **Constraint Satisfaction Rate**: fraction of generated assignments satisfying all known constraints.
- **Label Accuracy**: agreement between $\hat{Z}$ and true Z in synthetic tasks.
- **Central Effect Error**: error between the true-label treatment effect and the agent's central estimate.
- **LU-Bound Errors**: $|\hat{\Delta}_L - \Delta_L|$, $|\hat{\Delta}_U - \Delta_U|$.
- **Interval Coverage**: whether the reported interval contains the true-label treatment effect.
- **Invalid Output Rate**: frequency of non-binary labels, impossible counts, broken code, or inconsistent summaries.

- **Runtime**: time required to produce a valid reconstruction.

For small synthetic datasets, Δ_L and Δ_U can be obtained by exhaustive enumeration, giving a true benchmark target. For larger datasets, high-quality optimized solutions can serve as reference bounds.

3. Adaptive Tail-Focused Sampling

This extension keeps the MAPLE-style feasible assignment framework but changes the sampling strategy. Instead of drawing assignments uniformly or randomly from $\mathcal{Z}$, the sampler deliberately targets underexplored parts of the treatment-effect distribution. The goal is to approximate both the bulk of the feasible distribution and its tails more efficiently.

Algorithm Sketch

1. Generate an initial MAPLE-style Monte Carlo sample.
2. Compute $\Delta_Z(\tau)$ for all assignments.
3. Divide the observed treatment-effect range into bins.
4. Identify sparse bins, especially in the lower and upper tails.
5. Start new chains from assignments located near sparse bins.
6. Use feasible label swaps to move locally within $\mathcal{Z}$.
7. Prefer moves that enter underrepresented bins or expand the observed range.
8. Continue until histogram shape and tail estimates stabilize.

This approach is different from pure optimization. It does not only want the minimum and maximum; it wants a better map of the feasible treatment-effect distribution, especially rare regions.

Possible Outputs

- Tail-enriched treatment-effect histogram.
- Estimated 1st and 99th percentiles.
- Estimated minimum and maximum.
- Number of newly discovered tail assignments.
- Comparison against ordinary MAPLE Monte Carlo.

This could be positioned as a middle ground between MAPLE sampling and extremal optimization.

15. Joint-Covariate Extensions

1. Copula-Based Joint Covariate Reconstruction

The joint-covariate problem is to reconstruct $X_i = (X_{i1}, \ldots, X_{ip})$ for each patient using only published marginal summaries and limited dependency information. Independent reconstruction can match each marginal distribution but may create unrealistic patient profiles. For example, it may break relationships between age, ECOG status, metastatic disease, biomarker status, and baseline prognosis. A copula approach separates marginal distributions from dependence.

Workflow

1. Reconstruct each covariate marginal from published summaries.
2. Convert covariates to latent uniform scores using their marginal CDFs.
3. Use a copula to model dependence among covariates.
4. Generate synthetic joint covariate matrices.
5. Transform latent values back to observed covariate scales.
6. Repair or reweight samples to match published constraints.
7. Search over copula parameters to find central and edge-case joint reconstructions.

Gaussian Copula Version For a Gaussian copula, sample latent variables U_i through $Y_i \sim N(0, R)$, where R is a correlation matrix, then set $U_{ij} = \Phi(Y_{ij})$. Each U_{ij} is transformed into the observed covariate scale using the reconstructed marginal distribution of covariate j. The main unknown is R. If published pairwise relationships are unavailable, R can be varied over plausible values to study edge cases.

Edge-Case Search Edge cases are generated by searching over dependence structures, not just covariate values.

Examples include:

- strongest plausible association between poor ECOG and biomarker positivity,
- weakest plausible association between metastatic disease and biomarker positivity,
- maximum imbalance in high-risk patients across treatment arms while preserving published marginals,
- minimum and maximum feasible joint risk concentration.

The key output is a set of joint covariate datasets that satisfy published marginals while representing different plausible dependency structures.

TabDiff-Style Joint Covariate Reconstruction

TabDiff is a mixed-type diffusion model for tabular data generation. This makes it a natural candidate for joint covariate reconstruction because clinical covariates are usually heterogeneous, including binary, categorical, ordinal, and continuous variables.

In this project, a TabDiff-style model could be used to generate synthetic joint covariate matrices consistent with published clinical-trial summaries. The model would learn dependencies among covariates such as age, ECOG status, biomarker status, stage, metastatic disease, and treatment arm.

The basic workflow would be:

1. Train or fine-tune a mixed-type tabular diffusion model on external IPD or simulated trial datasets.
2. Condition generation on published trial-level summaries.
3. Generate many candidate joint covariate datasets.
4. Check each generated dataset against published constraints.
5. Reject, repair, or reweight generated datasets that violate constraints.
6. Search among accepted datasets for central and edge-case joint reconstructions.

The advantage of this approach is that TabDiff is designed to handle mixed-type tabular data and capture cross-column dependencies. This directly addresses the weakness of independent covariate reconstruction.

The limitation is that diffusion generation does not automatically guarantee exact satisfaction of published constraints. Therefore, TabDiff would likely need to be combined with constraint enforcement, such as rejection sampling, calibration, or optimization-based repair.

16. Meta Analysis for S(t) (MARS extension)

16.1. Feasible Survival Bands

Classical MARS produces a single pooled survival estimate by combining reconstructed IPD from multiple studies. However, reconstruction is generally not unique. Multiple patient-level datasets may satisfy the same Kaplan-Meier curves, numbers-at-risk tables, and published subgroup summaries.

A natural extension is to construct a *feasible survival band* rather than a single pooled curve. The workflow sketch follows:

1. Obtain reconstructed approximate IPD for each study.
2. Define reconstruction constraints by published summaries.
3. Generate multiple feasible reconstructions for each study.

4. Compute a pooled survival estimate for each feasible reconstruction.
5. Construct pointwise lower and upper envelopes: $S_L(t)$ and $S_U(t)$

The resulting band represents the range of pooled survival curves consistent with published information. This shifts the focus from obtaining a single reconstructed estimate to characterizing the collection of plausible survival trajectories.

Two-Stage Feasible Band Procedure A practical implementation is to use a two-stage procedure. The first stage estimates the distribution of feasible pooled survival curves using Monte Carlo reconstruction. The second stage uses constrained optimization to search for boundary curves that may be missed by random sampling.

Stage 1: Feasible Distribution Estimation

Generate M feasible reconstruction sets. Each set contains one feasible reconstructed IPD dataset for every study in the meta-analysis.

For reconstruction set m, compute the study-level survival curves $S_1^{(m)}(t), \dots, S_K^{(m)}(t)$, then pool them using the MARS procedure to obtain $S_{\text{pool}}^{(m)}(t)$.

The resulting collection $\{S_{\text{pool}}^{(1)}(t), \dots, S_{\text{pool}}^{(M)}(t)\}$ estimates the feasible distribution of pooled survival curves induced by reconstruction uncertainty.

Pointwise empirical summaries can be computed as $Q_p(t) = \text{Quantile}_p\{S_{\text{pool}}^{(1)}(t), \dots, S_{\text{pool}}^{(M)}(t)\}$, where $p \in \{0.05, 0.50, 0.95\}$.

Stage 2: Boundary Estimation

Monte Carlo samples may miss rare boundary reconstructions. Therefore, after estimating the central feasible distribution, run constrained optimization to identify lower and upper survival envelopes.

For each clinically relevant time point t^*, solve two optimization problems:

$$S_L(t^*) = \min_{D \in \mathcal{D}} S_{\text{pool}}(t^*; D), \qquad S_U(t^*) = \max_{D \in \mathcal{D}} S_{\text{pool}}(t^*; D),$$

where $\mathcal{D}$ is the set of reconstructed datasets satisfying all published constraints.

The optimization can be initialized from the lower-tail and upper-tail Monte Carlo reconstructions. Local moves modify reconstructed patient-level assignments while preserving numbers-at-risk, event counts, subgroup counts, and other published constraints. Candidate moves are accepted only if feasibility is preserved.

Algorithm Sketch

1. Input reconstructed IPD candidates, published constraints, and a time grid $\mathcal{T} = \{t_1, \dots, t_J\}$.

2. Generate M feasible reconstruction sets $D^{(1)}, \ldots, D^{(M)}$.

3. For each m, compute the pooled MARS survival curve $S_{\text{pool}}^{(m)}(t)$.

4. Estimate the central feasible distribution using empirical quantiles $Q_p(t)$.

5. For each $t_j \in \mathcal{T}$, select lower-tail and upper-tail Monte Carlo reconstructions as optimization starting points.

6. Run constrained minimization of $S_{\text{pool}}(t_j)$ to estimate $S_L(t_j)$.

7. Run constrained maximization of $S_{\text{pool}}(t_j)$ to estimate $S_U(t_j)$.

8. Repeat across all $t_j \in \mathcal{T}$.

9. Return the median feasible curve $Q_{0.50}(t)$, empirical uncertainty bands $Q_{0.05}(t), Q_{0.95}(t)$, and optimized feasible envelope $[S_L(t), S_U(t)]$.

This separates two goals: Monte Carlo reconstruction describes the typical feasible survival behavior, while constrained optimization estimates the most optimistic and pessimistic pooled survival curves still compatible with the published evidence.

Tail-Enriched Feasible Sampling

Instead of using constrained optimization only to find endpoints, we can use a tail-enriched sampling strategy. The goal is to preserve the Monte Carlo estimate of the central feasible distribution while adding targeted exploration of rare lower-tail and upper-tail reconstructions.

The procedure begins with an ordinary Monte Carlo sample $D^{(1)}, \ldots, D^{(M)}$, producing pooled curves $S_{\text{pool}}^{(1)}(t), \ldots, S_{\text{pool}}^{(M)}(t)$. These samples define the central feasible distribution.

Next, select reconstructions from the lower and upper empirical tails. For a fixed time point t^*, define lower-tail seeds as reconstructions with $S_{\text{pool}}(t^*) \leq Q_{0.05}(t^*)$ and upper-tail seeds as reconstructions with $S_{\text{pool}}(t^*) \geq Q_{0.95}(t^*)$.

From these seeds, run constraint-preserving local sampling chains. Moves are allowed only if they preserve the published reconstruction constraints. The acceptance rule is tilted toward the target tail. For lower-tail exploration, moves decreasing $S_{\text{pool}}(t^*)$ are favored. For upper-tail exploration, moves increasing $S_{\text{pool}}(t^*)$ are favored.

The final output keeps two objects separate:

- the ordinary Monte Carlo distribution, used to describe the central feasible region;
- the tail-enriched sample, used to describe rare feasible boundary behavior.

This avoids distorting the central distribution while still improving exploration of low-probability feasible tails.

16.2. Bayesian Reconstruction Meta Analysis

Reconstruction uncertainty can be represented probabilistically. In this framework, each feasible reconstruction is viewed as a draw from a posterior distribution over patient-level datasets.

The workflow would proceed as follows:

1. Generate a large collection of feasible reconstructions.
2. Compute pooled survival estimates for each sample.
3. Aggregate the resulting collection of survival curves.

The primary object of interest becomes a posterior distribution over survival functions,

$$\{S^{(1)}(t), \dots, S^{(M)}(t)\}.$$

From this collection one can obtain posterior means, pointwise credible intervals, median survival distributions, or other functionals of interest.

Bayesian Bootstrap Specification

A simple probabilistic version can be built from the reconstructed study-level survival curves without specifying a full parametric survival model.

Suppose study k has feasible reconstructed survival curves $S_k^{(1)}(t), \dots, S_k^{(M_k)}(t)$. Instead of assigning equal weight to every reconstruction, draw Bayesian bootstrap weights

$$\omega_k^{(1)}, \dots, \omega_k^{(M_k)} \sim Dirichlet(1, \dots, 1).$$

For each posterior draw b, sample or weight one feasible reconstruction from each study, then compute the pooled curve

$$S_{\text{pool}}^{(b)}(t) = \sum_{k=1}^{K} v_k^{(b)} S_k^{(b)}(t),$$

where $v_k^{(b)}$ are study-level pooling weights, such as inverse-variance weights, sample-size weights, or Bayesian bootstrap weights over studies.

The procedure is:

1. Generate feasible reconstructions for each study.
2. Draw reconstruction-level Bayesian bootstrap weights within each study.
3. Draw study-level Bayesian bootstrap weights across studies.
4. Compute a pooled survival curve for each posterior draw.

5. Summarize $\{S^{(1)}_{\text{pool}}(t), \ldots, S^{(B)}_{\text{pool}}(t)\}$ using posterior means and pointwise credible bands.

This approach treats uncertainty as arising from two sources: uncertainty over feasible reconstructions within each study and uncertainty over how studies contribute to the pooled curve.

Hierarchical Bayesian Specification

A more structured model treats each study survival curve as a noisy realization of an underlying pooled survival function.

Represent each reconstructed survival curve on a common time grid $\mathcal{T} = \{t_1, \ldots, t_J\}$. Let $Y^{(m)}_{kj}$ denote a transformed survival estimate from study k, reconstruction m, at time t_j. A useful transformation is the complementary log-log scale,

$$Y^{(m)}_{kj} = \log\{-\log S^{(m)}_k(t_j)\},$$

which maps survival probabilities from $(0, 1)$ to the real line.

A hierarchical model can be specified as

$$Y^{(m)}_{kj} = \mu_j + b_{kj} + \epsilon^{(m)}_{kj},$$

where μ_j is the pooled survival trajectory on the transformed scale, b_{kj} is a study-specific deviation, and $\epsilon^{(m)}_{kj}$ captures reconstruction-level variation.

One possible prior structure is

$$b_{kj} \sim N(0, \tau^2_j), \qquad \epsilon^{(m)}_{kj} \sim N(0, \sigma^2_{kj}),$$

with smoothness priors imposed on μ_j, such as a random-walk prior,

$$\mu_j \sim N(\mu_{j-1}, \eta^2).$$

After posterior sampling, transform back to the survival scale using

$$S_{\text{pool}}(t_j) = \exp\{-\exp(\mu_j)\}.$$

The procedure is:

1. Generate multiple feasible reconstructions for each study.

2. Evaluate each reconstructed survival curve on a common time grid.

3. Transform survival probabilities to the complementary log-log scale.

4. Fit a hierarchical model with study-level and reconstruction-level variation.
5. Transform posterior draws of μ_j back to obtain posterior pooled survival curves.
6. Summarize posterior survival trajectories using means, credible bands, and derived quantities such as median survival or RMST.

This model separates pooled survival behavior, between-study heterogeneity, and within-study reconstruction uncertainty.

16.3. Functional Meta Analysis of Entire Curves

Most survival meta-analyses reduce each study to a low-dimensional summary such as a HR, median survival time, or RMST. An alternative approach is to treat each reconstructed survival curve as a functional observation.

Under this perspective, the observed data become

$$S_1(t), S_2(t), \ldots, S_K(t),$$

where each study contributes an entire survival trajectory rather than a single scalar summary.

A potential workflow is:

1. Reconstruct survival curves for all studies.
2. Represent curves on a common time grid.
3. Treat each survival curve as a functional observation.
4. Learn common patterns across studies.
5. Construct pooled functional survival estimates.

Functional Principal Component Analysis

Functional principal component analysis (FPCA) provides a dimension reduction framework for collections of survival curves. After reconstructing study-level survival functions and evaluating them on a common time grid, each study contributes a functional observation $S_k(t)$.

The first step is to estimate the mean survival curve,

$$\mu(t) = \frac{1}{K} \sum_{k=1}^{K} S_k(t),$$

and then decompose the residual variation around the mean into orthogonal modes of variation:

$$S_k(t) = \mu(t) + \sum_{r=1}^{R} \xi_{kr}\phi_r(t).$$

Here, $\phi_r(t)$ denotes the r-th functional principal component and ξ_{kr} is the corresponding score for study k.

The resulting components provide an interpretable representation of how survival trajectories differ across studies. For example, one component may represent early mortality differences, while another captures long-term survivor effects. The corresponding scores can be used to visualize study similarities, identify outlying studies, and quantify dominant sources of variation within the literature.

A possible workflow is:

1. Reconstruct study-level survival curves.
2. Evaluate all curves on a common time grid.
3. Estimate the mean survival function.
4. Compute functional principal components.
5. Analyze component scores and explained variance.
6. Reconstruct pooled or representative survival trajectories using a small number of principal components.

Rather than producing only a pooled survival estimate, FPCA provides a low-dimensional representation of the major survival patterns observed across studies.

Functional Mixed-Effects Models

A second approach is to explicitly model study-level variation using functional random effects. Rather than assuming that all studies arise from a single common survival trajectory, each study is allowed to deviate from a pooled mean function.

One possible specification is

$$S_k(t) = \mu(t) + b_k(t) + \epsilon_k(t),$$

where $\mu(t)$ represents the pooled survival trajectory, $b_k(t)$ is a study-specific functional random effect, and $\epsilon_k(t)$ represents residual variation.

The pooled function captures the average survival behavior across studies, while the random effects quantify between-study heterogeneity throughout the follow-up period. Unlike traditional random-effects meta-analysis, heterogeneity is modeled as a function of time rather than a single scalar quantity.

The workflow would proceed as follows:

1. Reconstruct study-level survival curves.
2. Represent curves on a common time grid.
3. Estimate a pooled mean survival function.
4. Fit study-specific functional random effects.
5. Quantify heterogeneity across follow-up.
6. Construct pooled survival estimates together with study-level deviations.

An important output is a time-varying characterization of heterogeneity. Some regions of the survival curve may exhibit strong agreement across studies, while others may show substantial divergence. Consequently, functional mixed-effects models provide a direct framework for studying when and where survival behavior differs throughout follow-up.

16.4. Survival Curve Heterogeneity Maps

Traditional meta-analysis often summarizes between-study heterogeneity using a single statistic such as I^2. However, disagreement between studies is unlikely to remain constant over the entire follow-up period. Studies may exhibit strong agreement at early time points while diverging substantially at later horizons due to differences in patient populations, follow-up duration, treatment mechanisms, or censoring patterns.

A natural extension of MARS is to estimate heterogeneity as a function of time rather than as a single scalar quantity. The workflow would proceed as follows:

1. Reconstruct study-level survival curves.
2. Estimate a pooled survival function using MARS.
3. Compute between-study variability at each time point.
4. Construct a time-varying heterogeneity measure:
   - variance of $S(t)$,
   - standard deviation of $S(t)$,
   - functional analogues of I^2.
5. Visualize heterogeneity as a function of time.

The primary output is a heterogeneity map showing where studies exhibit strong agreement and where substantial disagreement emerges. Such maps may reveal that treatment effects are highly consistent during early follow-up but become increasingly heterogeneous among long-term survivors. This framework shifts the focus from estimating a single pooled survival curve toward understanding how agreement between studies evolves throughout the survival trajectory.

16.5. Change-Point MARS

Many survival curves exhibit distinct phases of behavior. Examples include rapid early mortality followed by stabilization, delayed treatment benefits due to immunotherapy, or long-term survivor effects that emerge only after extended follow-up. Standard meta-analysis typically ignores these transitions and treats the entire survival trajectory as a single object.

A change-point extension of MARS would explicitly identify locations where survival behavior changes. The workflow would proceed as follows:

1. Reconstruct survival curves for all studies.
2. Estimate study-specific change points using segmented survival models or piecewise hazard models.
3. Identify common change-point locations across studies.
4. Partition survival trajectories into distinct temporal phases.
5. Perform separate pooling within each phase.

Potential change points include:

- onset of treatment benefit,
- transition to long-term survivor populations,
- periods of increased mortality,
- regions where survival curves begin to separate.

The resulting meta-analysis would characterize not only the pooled survival function but also the temporal structure of treatment response. Rather than viewing survival as a single continuous process, the framework identifies clinically meaningful stages where the underlying dynamics may change substantially across studies.

16.6. Multi-Agent Consensus Reconstruction

Many reconstruction choices involve ambiguity, including curve digitization, coordinate extraction, and handling incomplete reporting. Instead of relying on a single reconstruction, multiple agents can independently generate candidate reconstructions.

The workflow would proceed as follows:

1. Multiple agents reconstruct the same study.
2. Each agent produces reconstructed IPD and survival estimates.
3. Agreement between agents is assessed.
4. Consensus reconstructions are formed through aggregation procedures.

5. Meta-analysis is performed using consensus outputs.

This approach is analogous to ensemble learning and may improve robustness when published information is incomplete or ambiguous.

17. 6/19 Meeting Notes

Flesh out one or two promising ideas (pipelines) and summarize results. Rather than mentalizing MAPLE or MARS extension, it is acceptable to start from scratch if needed. most promising is

Survival Curve Heterogeneity Maps

TabDiff for Reconstruction

"Wow, It really works!"